Grouping Structure of Linear Partial Differential Equations and Its Use in Learning Them from Data

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Abstract. In Fourier variables a linear partial differential equation with constant coefficients is a polynomial, and a factorization of this polynomial splits every solution into groups of modes, one group per factor. We study this grouping structure and use it to learn equations from noisy data. On the analytical side we show that the groups are the factors of the symbol and not the additive terms of the operator, that the split is unique up to a finite-dimensional overlap given by an intersection number, that its stability is governed by the conditioning of the sum of the factors' solution spaces (for two factors, their Friedrichs angle), and that without a priori bounds finite data can support but never certify an equation. On the algorithmic side we group twice: Fourier modes into factors within one sample of the field, by fitting the curves of the dispersion relation, and factors across posterior samples, by Dirichlet-process clustering. The output is a distribution over equations with intervals and labels stating what the data support. On twenty random equations not used in development the method returns the right or an equivalent equation on nineteen, fourteen of them with a confident label, and is never wrong without warning, while each of the three library-regression baselines run on this set is wrong without warning on at least thirteen. On sixty grid inputs with independent fields and noise it is always right, and on the ten-equation development benchmark, in three noise and sampling designs, it recovers all thirty factor sets.

Key words. equation discovery, partial differential equations, operator factorization, Friedrichs angle, matrix pencil, Dirichlet process, uncertainty quantification

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1 Introduction

A partial differential equation with constant coefficients becomes, in Fourier variables, a polynomial: the operator $(\partial_t + c\,\partial_x)(\partial_t - \nu\,\partial_x^2)$ of a quantity that is carried by a flow and diffuses becomes $(\tau + c\,\xi)(\tau - \nu\,\xi^2)$. Polynomials can be multiplied and factored, and a factorization of the symbol splits every solution into pieces, one for each factor: here a travelling profile, whose Fourier modes lie on the line $\lambda = -ic\kappa$, and a diffusing one, whose modes lie on the parabola $\lambda = -\nu\kappa^2$ (Fig. 1). This paper is about that splitting, which we call the *grouping structure* of the equation, and about how a machine-learning algorithm can find it in noisy data and use it to recover the equation. The example matters in practice: methods that regress u_t on a library of terms look for a single sum of terms, and every one we tested returns a wrong equation for it at 5% noise, without warning.

Two questions organize the work. The first is analytical: which pieces of an equation does a solution actually carry, how uniquely, and how stably? The answer is that the pieces are the

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(a) oscillation (b) damping

Figure 1: The idea of the paper on the equation $(\partial_t + c\,\partial_x)(\partial_t - \nu\,\partial_x^2)u = 0$. Every Fourier mode $e^{\lambda t + i\kappa x}$ of a solution lies on the zero set of one factor of the symbol $(\tau + c\xi)(\tau - \nu\xi^2)$: circles on the line of the transport factor, squares on the parabola of the diffusion factor (schematic; Fig. 5 shows the modes found in noisy data). Learning the equation means learning these two groups.

factors of the symbol, not the additive *terms* of the operator; that the split is unique up to a finite-dimensional overlap whose dimension is an intersection number; and that its stability is governed by the Friedrichs angle between the solution spaces of the factors. The second question is algorithmic: how can one learn the groups from finitely many noisy observations, and what should the output say about how well they were learned? Our answer is an algorithm that groups twice. Within one sample of the field it groups Fourier modes into factors, using the fact that the modes of one factor lie on one curve of the dispersion relation. Across posterior samples it groups the factors found in each sample into clusters, using a Dirichlet-process mixture. The output is a distribution over equations with labels that say what the data can and cannot support.

Contributions

1. An analytical account of the grouping structure: identification and uniqueness (Theorem 2.9 and Proposition 2.10), the factor decomposition with its exact ambiguity $\delta(P)$ (Theorem 2.14), its conditioning (Theorem 2.17), and the impossibility of grouping by terms (Propositions 2.16 and 2.19).
2. An algorithm that turns this structure into a learning procedure: interpolation with a sampling-based treatment of the high-frequency band, a matrix pencil per wavenumber, a joint grouping-and-fitting of the modes, and a Bayesian layer that groups factors across samples and attaches intervals whose empirical coverage is 0.94 at nominal 0.90.
3. Evidence: on twenty random equations not used in development the method is right or equivalent on 19 (14 with a confident label, 5 with an abstaining label and correct factors) and never wrong without warning, against 13 or more such failures for each of the three library-regression baselines run on this set; on 60 grid inputs with independent fields and noise it is always right, against 13 silent failures for weak SINDy; on a benchmark of ten equations in three designs, on which its parameters were tuned, it recovers all 30 factor sets, while the best library regression is right or equivalent on 25 and wrong without warning on 5.

Relation to prior work Most methods that learn a PDE from data regress a time derivative on a library of candidate terms: SINDy and PDE-FIND [2, 17], their weak, ensemble and Bayesian variants [13, 6, 8], and Gaussian-process formulations of linear operators [16]. Their unit is the *term*. Dynamic mode decomposition [19, 18] works with eigenvalues, as we do, but returns a linear operator on the data, not an equation, and does not group eigenvalues by the factor they belong to. Exponential fitting by matrix pencils [10] is a classical tool of signal processing; the algebra of constant-coefficient operators, the Ehrenpreis–Palamodov fundamental principle and the solvability results behind Theorem 2.14 are classical analysis [5, 15, 12, 9, 14]; Dirichlet-process mixtures are standard Bayesian clustering [1]. What is new here is the connection between them. (i) We identify the *factor* of the symbol, not the term, as the unit a learning method should group by, and we prove that, for two terms whose varieties meet in finitely many points, per-term grouping is impossible outside a finite-dimensional set of solutions (Propositions 2.16 and 2.19). (ii) We state what finite data can and cannot certify about such groups (Theorem 2.9 and Proposition 2.13) and give the exact conditioning of the separation between groups (Theorem 2.17 and Proposition 5.1). (iii) We build an algorithm that groups twice, modes into factors within a sample and factors across posterior samples, and whose output carries these limits as labels and intervals.

Scope Everything proved and implemented concerns scalar linear equations with constant coefficients in one time and one space variable, on rectangular windows. Section 3 explains which parts of the structure are lost for nonlinear equations, and Remark 4.2 how the spectral step reads variable coefficients; neither is claimed as a result.

Organization Section 2 develops the analytical framework: the operator ring, exponential-polynomials, identification, the factor decomposition and its spectral projectors, the Friedrichs angle, and the interaction term with its energy. Section 3 discusses nonlinear equations. Section 4 describes the deterministic algorithm and Section 5 its statistical layer. Section 6 follows one data set through every stage and reports benchmarks. Section 7 separates the inference problems that have a remedy from those that do not, and Section 8 concludes. Figure 2 summarizes the method; proofs are in Appendix A. *Reader's guide.* The algorithm uses four results: the factor decomposition (Theorem 2.14), the stability of the split (Theorem 2.17), the counting condition (Proposition 4.1) and the limits of finite data (Proposition 2.13), which explain the labels. A reader interested mainly in the method can skim the material on exponential-polynomials and identification on a first reading.

2 The analytical framework

This section develops the grouping structure in the order in which the algorithm will use it: derivatives become polynomials (Section 2.1); a solution carries an ideal of equations, and exponential-polynomials are exactly the solutions that carry too little (Sections 2.2 and 2.3); this decides when the equation can be recovered at all (Sections 2.4 and 2.5); the factors of the symbol split the solution, with projectors given by the eigenvalues (Section 2.6); the split is stable exactly when the Friedrichs angle is positive (Section 2.7); and inside one factor the additive terms interact (Section 2.8). Table 1 collects the notation.

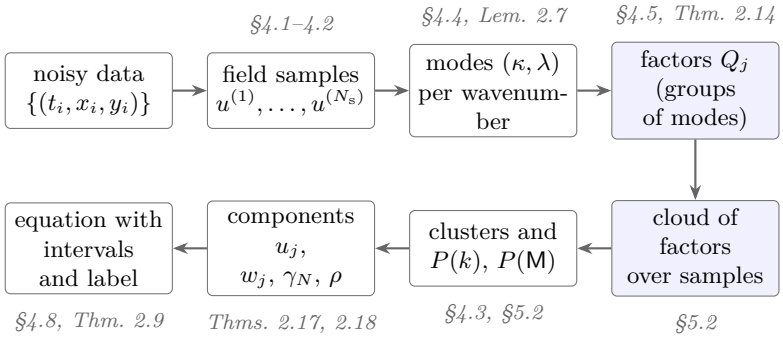


Figure 2: The method at a glance. Shaded: the two grouping steps, of modes into factors within one sample and of factors across samples. Each box names the section that describes it and the result that justifies it.

Table 1: Notation for the analysis. Each symbol has one meaning throughout the paper; Table 2 lists the notation of the algorithm.

Symbol	Meaning	Introduced
Operators and their algebra		
$\mathcal{R} = \mathbb{C}[\tau, \xi]$	ring of constant-coefficient operators; $\tau \leftrightarrow \partial_t$, $\xi \leftrightarrow \partial_x$	Definition 2.1
$P,\ Q_j,\ P_i,\ e_i$	an operator and its symbol; a factor; an irreducible factor and its multiplicity	Sections 2.1 and 2.6
$V(P)$	characteristic variety $\{P = 0\} \subset \mathbb{C}^2$	Definition 2.1
$\kappa,\ \lambda(\kappa)$	real wavenumber ($\xi = i\kappa$); temporal rate of a mode	Section 2.1
$\Lambda_j(\xi)$	branch of an evolution factor $Q_j = \tau - \Lambda_j(\xi)$	(6)
$\mathrm{Ann}(u)$	annihilator: all operators that kill u	Definition 2.3
$G_u,\ J_u,\ \Sigma_{\mathrm{p}}(u)$	intrinsic operator $\gcd \mathrm{Ann}(u)$; cofactor ideal; point spectrum $V(J_u)$	(2)
EP	exponential-polynomials	Definition 2.6
$N,\ \delta(P)$	number of factors; dimension of the overlap between their solution spaces	Theorem 2.14
$m_j,\ d_j$	degree of factor j in τ and in ξ	Section 4.3
Geometry and energy		
$\mathcal{S}$	sum operator $(u_1, \ldots, u_N) \mapsto \sum_j u_j$	Section 2.7
γ_N	conditioning constant of $\mathcal{S}$; error amplification $\gamma_N^{-1/2}$	(7)
$c_F,\ \theta_F$	Friedrichs cosine and angle; $\gamma_2 = 1 - c_F$	Section 2.7
H^s	Sobolev space of index s	Section 2.7
$K;\ \mathcal{A}_n,\ \Phi_n,\ \Delta_n$	number of additive terms of a factor; a term; its flow $e^{t\mathcal{A}_n}$; increment $\Phi_n - \mathrm{Id}$	Section 2.8
$\hat{\mathcal{A}}_n(\kappa)$	Fourier symbol of the term $\mathcal{A}_n$	(10)
$E_{\mathrm{int}},\ \rho$	interaction term; interaction ratio $\|E_{\mathrm{int}}\|/\|u(T) - u_0\|$	(9), (11)
w_j	energy share of group j	(11)

Table 2: Notation for the algorithm (continued from Table 1).

Symbol	Meaning	Introduced
$T,\ L$	length of the time window; spatial period	Section 4.1
$\Delta t,\ \Delta x;\ h$	grid spacings; mean spacing $\sqrt{TL/n}$ of n scattered points	Section 4.2
$\sigma_n,\ \sigma_\kappa$	observation noise standard deviation; its value in one Fourier bin	Sections 4.1 and 4.2
$\mathcal{K},\ z$	band of usable wavenumbers; detection threshold in units of σ_κ	Section 4.2
h_κ	spread $\max_{\ell,\ell'}\lvert\kappa_\ell - \kappa_{\ell'}\rvert$ of the wavenumbers on which a factor is fitted	Proposition 4.1
$A_p,\ \sigma_{\lambda,p}$	amplitude of mode p; standard deviation of its rate	Section 4.4
$N_\mathrm{s},\ u^{(q)},\ \pi_q$	number of posterior field samples; sample q; its weight	Section 4.1
M	a structure (number and degrees of the factors)	Section 4.3
B_j	orthonormal basis of the sampled kernel of Q_j	Section 4.7
E_j	kernel exponentials of Q_j evaluated at the observation points	Section 4.9
ω^2	prior variance of the component coefficients	(24)
$\eta_r,\ \ell^{(r)}$	amplitudes and length scales of the Gaussian-process kernel	(13)
$\eta,\ \eta_h,\ \pi_h,\ N_\eta$	hyperparameters; a draw of them; its weight; number of draws	(14)
$\beta,\ W$	coefficient vector of a factor; cloud of factors	(27)
α	concentration of the Dirichlet process	Section 5.2
$k,\ k_\mathrm{eff},\ \bar{w}_j$	number of clusters (groups) found; effective number of groups; cluster masses	Section 5.2
FIT-COMPATIBLE, ID-KER, MISFIT, UNDECIDED	output labels	Section 4.8

2.1 Derivatives become polynomials

Throughout, $\Omega \subset \mathbb{R}^2$ is a domain in the variables (t, x) and every operator has constant coefficients,

$$P(\partial_t, \partial_x) = \sum_{j,k} c_{jk}\, \partial_t^j \partial_x^k, \qquad c_{jk} \in \mathbb{C}.$$

The exponentials $\psi_{\tau,\xi}(t, x) = e^{\tau t + \xi x}$, $(\tau, \xi) \in \mathbb{C}^2$, are common eigenfunctions of all such operators: $\partial_t \psi_{\tau,\xi} = \tau \psi_{\tau,\xi}$, $\partial_x \psi_{\tau,\xi} = \xi \psi_{\tau,\xi}$ and therefore

$$P(\partial)\, \psi_{\tau,\xi} = P(\tau, \xi)\, \psi_{\tau,\xi}. \tag{1}$$

Differentiation is thus replaced by multiplication, and operators are replaced by polynomials.

Definition 2.1 (Operator ring, symbol, characteristic variety)**.** The constant-coefficient operators form the commutative ring $\mathcal{R} = \mathbb{C}[\partial_t, \partial_x] \cong \mathbb{C}[\tau, \xi]$. The polynomial $P(\tau, \xi)$ is the *symbol* of $P(\partial)$, and its zero set $V(P) = \{(\tau, \xi) \in \mathbb{C}^2 : P(\tau, \xi) = 0\}$ is the *characteristic variety*.

By (1), $\psi_{\tau,\xi}$ solves $P(\partial)u = 0$ exactly when $(\tau, \xi) \in V(P)$. A Fourier mode in space is the case $\xi = i\kappa$ with wavenumber $\kappa \in \mathbb{R}$; then $P(\tau, i\kappa) = 0$ is the *dispersion relation*, and the roots $\tau = \lambda(\kappa)$ are the temporal rates of the mode $e^{\lambda(\kappa)t + i\kappa x}$.

Convention We keep the algebraic variable ξ in statements about the ring and write $\xi = i\kappa$ only when we evaluate on Fourier modes. With this convention a symbol has real coefficients exactly when the equation is real, and no factor i ever appears inside $\mathcal{R}$.

Example 2.2 (Transport plus diffusion: the running example). The operator $P = (\partial_t + c\,\partial_x)(\partial_t - \nu\,\partial_x^2)$, with speed $c > 0$ and diffusivity $\nu > 0$, has symbol

$$P(\tau, \xi) = Q_1 Q_2, \qquad Q_1 = \tau + c\,\xi, \qquad Q_2 = \tau - \nu\,\xi^2.$$

On Fourier modes the two factors give two dispersion branches, $\lambda_1(\kappa) = -ic\kappa$ (a wave that travels without decay) and $\lambda_2(\kappa) = -\nu\kappa^2$ (a standing mode that decays). A solution is any superposition of modes on the two branches. The equation is one object; its *two factors* are two groups of modes. Recovering those groups from data is the subject of this paper. We use $c = 1$, $\nu = 0.3$ in all numerical illustrations.

2.2 What a solution knows: the annihilator

A single function satisfies many equations: e^x is annihilated by $\partial_x - 1$, by $\partial_x^2 - 1$ and by every multiple of $\partial_x - 1$.

Definition 2.3 (Annihilator)**.** For a distribution u on Ω, $\mathrm{Ann}(u) = \{Q \in \mathcal{R} :\ Q(\partial)u = 0\}$.

$\mathrm{Ann}(u)$ is an ideal: it is closed under sums and under multiplication by any operator, because $(SQ)(\partial)u = S(\partial)(Q(\partial)u) = 0$. It collects every constant-coefficient equation that u satisfies. We use three classical facts about $\mathbb{C}[\tau, \xi]$ [3]: unique factorization into irreducible polynomials; Bézout's theorem (two polynomials without a common factor have finitely many common zeros); and Hilbert's Nullstellensatz (if q vanishes on the zero set $V(I)$ of an ideal I, then $q^k \in I$ for some k). All three require the complex field, which is why symbols are always considered over $\mathbb{C}$.

Lemma 2.4 (Splitting off the greatest common divisor)**.** *Every nonzero ideal $I \subset \mathcal{R}$ factors as $I = g \cdot J$, where $g = \gcd(I)$ and J is an ideal with finite zero set $V(J)$.*

The proof (Appendix A) is elementary: divide by g, then use Bézout on two elements of J without a common factor. Applied to $I = \mathrm{Ann}(u)$ the lemma gives the decomposition

$$\mathrm{Ann}(u) = G_u\, J_u, \qquad G_u = \gcd \mathrm{Ann}(u), \qquad \Sigma_{\mathrm{p}}(u) = V(J_u) \text{ finite.} \tag{2}$$

The polynomial G_u is the part of the equation that u fills out along whole curves of $\mathbb{C}^2$; the finite set $\Sigma_{\mathrm{p}}(u)$, the *point spectrum* of u, lists the isolated exponentials that u happens to contain. We call G_u the *intrinsic operator* of u: it depends on u alone, whereas an annihilator P supplied from outside may contain spurious factors.

Reading the two pieces The two objects in (2) answer different questions. The polynomial G_u is a differential equation: its zero set $V(G_u)$ is a union of curves in $\mathbb{C}^2$, and each curve is a continuous family of exponential modes $e^{\tau t+\xi x}$, all solving $G_u(\partial)v = 0$. The point spectrum $\Sigma_{\mathrm{p}}(u)$ is a list: the finitely many individual exponential modes that are present in this particular function u and are not accounted for by G_u. In short, G_u identifies an equation that describes a whole family of modes; $\Sigma_{\mathrm{p}}(u)$ identifies the isolated modes that u happens to contain. Example 2.5 shows the two extreme cases.

Example 2.5 (Two extremes). For $u = e^x$, $\mathrm{Ann}(u) = (\tau,\ \xi - 1)$ is the ideal of the single point $(0, 1)$; hence $G_u = 1$ and $\Sigma_{\mathrm{p}}(u) = \{(0, 1)\}$: the function knows one exponential and no equation. For $u = f(x - ct)$ with a profile f that is not a finite sum of exponentials times polynomials, $\mathrm{Ann}(u) = (\tau + c\xi)$; hence $G_u = \tau + c\xi$ and $\Sigma_{\mathrm{p}}(u) = \emptyset$: the function knows its equation completely (Theorem 2.9).

2.3 Exponential-polynomials: the solutions that remember too little

Definition 2.6 (Exponential-polynomials)**.** EP is the set of finite sums $\sum_j p_j(t, x)\, e^{\tau_j t+\xi_j x}$ with polynomials p_j and exponents $(\tau_j, \xi_j) \in \mathbb{C}^2$. $\mathrm{EP}(\Omega)$ denotes their restrictions to Ω.

For a function w let $\mathcal{R}w = \mathrm{span}\{\partial_t^j \partial_x^k w :\ j, k \geq 0\}$ be the space of all its derivatives. The map $Q \mapsto Q(\partial)w$ is onto $\mathcal{R}w$ with kernel $\mathrm{Ann}(w)$, so $\mathcal{R}w \cong \mathcal{R}/\,\mathrm{Ann}(w)$ as vector spaces.

Lemma 2.7 (Finite-dimension criterion)**.** *Let Ω be connected. Then $w \in \mathrm{EP}(\Omega)$ if and only if $\dim_{\mathbb{C}} \mathcal{R}w < \infty$. In that case $V(\mathrm{Ann}\, w)$ is the finite set of exponents $\{(\tau_j, \xi_j)\}$ that occur in w.*

Proof idea. We show that w is an exponential-polynomial if and only if its derivatives span a finite-dimensional space. ($\Rightarrow$) For one term $w = p(t, x)\, e^{at+bx}$,

$$\partial_t w = (p_t + a\, p)\, e^{at+bx}, \qquad \partial_x w = (p_x + b\, p)\, e^{at+bx}.$$

Differentiation never raises the degree of p, so every derivative of w lies in the finite-dimensional space of polynomials of degree at most $\deg p$ times e^{at+bx}; a finite sum of such terms stays in a finite sum of such spaces. ($\Leftarrow$) If $\mathcal{R}w$ is finite-dimensional, ∂_t and ∂_x map it to itself and act on it as two commuting matrices A and B. Their minimal polynomials m_A and m_B give two constant-coefficient differential equations,

$$m_A(\partial_t)\, w = 0, \qquad m_B(\partial_x)\, w = 0,$$

one in each variable. On every rectangle inside Ω their common solutions are exponential-polynomials, and connectedness extends this representation to all of Ω. The joint eigenvalues (a, b) of A and B are the exponents that occur in w. Details are in Appendix A. □

In brief. Deciding whether $w \in \mathrm{EP}$ is a finite-dimensional linear-algebra question: compute the rank of the derivatives of w. The joint eigenvalues of the two matrices are the exponents, that is, the points of $\mathbb{C}^2$ at which w lives. This is the computational core of everything that follows: the matrix pencil of Section 4.4 is exactly this eigenvalue computation, done for ∂_t one wavenumber at a time.

Example 2.8 (Rich and poor solutions of the heat equation). A single heat mode $e^{-\nu\kappa^2 t+i\kappa x}$ and any finite sum of such modes belong to EP. The heat kernel $t^{-1/2}e^{-x^2/(4\nu t)}$ does not: its x-derivatives are polynomials of increasing degree times the same Gaussian, so they span an infinite-dimensional space.

2.4 When does a solution determine its equation?

Let Ω be open and connected (on a disconnected set the pieces may obey different equations), let $u \in \mathcal{D}'(\Omega)$ satisfy $P(\partial)u = 0$, and let $P = \prod_i P_i^{e_i}$ (up to a nonzero constant factor) be the factorization of P into pairwise non-associate irreducible polynomials P_i with multiplicities e_i. The question is whether $\mathrm{Ann}(u) = (P)$: then every constant-coefficient equation satisfied by u is a consequence of P, and P is determined by u.

Theorem 2.9 (Identification criterion). *Put $v_i = (P/P_i)(\partial)\, u$ and $v_i' = (P/P_i^{e_i})(\partial)\, u$.*

(a) $\mathrm{Ann}(u) = (P)$ *if and only if* $v_i \notin \mathrm{EP}(\Omega)$ *for every* i.

(b) $V(\mathrm{Ann}\, u) = V(P)$ *if and only if* $v_i' \notin \mathrm{EP}(\Omega)$ *for every* i.

(c) If P is square-free, (a) and (b) coincide. If P is monic in τ of degree m, then $\mathrm{Ann}(u) = (P)$ *if and only if the only* $(\varphi_0, \dots, \varphi_{m-1}) \in \mathbb{C}[\xi]^m$ *with* $\sum_{j<m} \varphi_j(\partial_x)\, \partial_t^j u = 0$ *is zero.*

The idea is to strip one irreducible factor and look at what is left. The function v_i solves $P_i(\partial)v_i = 0$. If v_i is a generic solution of that factor, no smaller operator kills u. If v_i is an exponential-polynomial, some h coprime to P_i kills it, and then $h \cdot P/P_i$ is an annihilator of u that is not a multiple of P. The converse uses Lemma 2.4; the full proof is in Appendix A.

Proposition 2.10 (Uniqueness and the intrinsic operator). *(a) If* $\mathrm{Ann}(u) = (P)$*, then P is determined by u up to a nonzero constant and is the unique annihilator of minimal degree.*

(b) $\mathrm{Ann}(u) = (P)$ *if and only if* $\Sigma_{\mathrm{p}}(u) = \emptyset$ *and* $G_u \sim P$ *(equality up to a constant).*

(c) For any annihilator P, the multiplicity of P_i in G_u equals $e_i - \max\{k \le e_i : (P/P_i^k)(\partial)u \in \mathrm{EP}\}$. *Hence G_u is computed from any annihilator by at most* $\sum_i e_i$ *membership tests "*$\in \mathrm{EP}$*".*

Example 2.11 (The heat equation, continued). For $P = \tau - \nu\xi^2$ (irreducible) the criterion reads: $\mathrm{Ann}(u) = (P)$ if and only if $u \notin \mathrm{EP}$. One mode $e^{-\nu\kappa^2 t+i\kappa x}$ has $\mathrm{Ann}(u) = (\tau + \nu\kappa^2,\ \xi - i\kappa)$, a point of $V(P)$; it cannot distinguish the heat equation from the pure decay $\partial_t + \nu\kappa^2$, or from any other equation whose variety passes through the point $(-\nu\kappa^2, i\kappa)$. The heat kernel is not in EP, so every constant-coefficient equation it satisfies is a multiple of $\partial_t - \nu\partial_x^2$.

Remark 2.12 (Curves and points). The two cases of Example 2.11 sit at the two ends of the decomposition (2). A finite sum of modes $u \in \mathrm{EP}$ has an annihilator with finitely many zeros, so

$$u \in \mathrm{EP}: \qquad G_u = 1, \qquad \Sigma_{\mathrm{p}}(u) = \{(\tau_j, \xi_j)\} \text{ finite (its exponents)},$$

whereas the zero set of a nonconstant polynomial in two variables is a whole algebraic curve. For the heat kernel the situation is reversed:

$$u = t^{-1/2} e^{-x^2/(4\nu t)}: \qquad G_u = \tau - \nu\xi^2, \quad V(G_u) \text{ a curve}, \qquad \Sigma_{\mathrm{p}}(u) = \emptyset.$$

By Proposition 2.10(b) it is exactly this pair, an empty point spectrum together with $G_u \sim P$, that gives $\mathrm{Ann}(u) = (P)$: the solution fills the whole curve of its equation and contains no isolated mode besides.

Role in the algorithm. Data never contain the heat kernel, they contain finitely many sampled modes; a finite set of modes is always an exponential-polynomial. The algorithm therefore never claims that an equation is *identified*. It asks whether the modes it sees are *compatible* with a factored symbol and are rich enough in each factor (enough distinct wavenumbers per branch) for the factor to be pinned down within its class. This is the meaning of the label FIT-COMPATIBLE in Section 4.8.

2.5 What finite data can and cannot certify

Theorem 2.9 speaks about u, not about finitely many numbers. Two facts frame what an algorithm can promise.

Recall that $\mathcal{R}u = \mathrm{span}\{\partial_t^j \partial_x^k u : \ j, k \geq 0\}$ is the vector space generated by the derivatives of u (Section 2.3). The *jet* of order M of u at a point $z_0 = (t_0, x_0)$ is the list of its derivatives of order at most M at that point; for example

$$j^2_{z_0}(u) = (u, \ u_t, \ u_x, \ u_{tt}, \ u_{tx}, \ u_{xx})\big|_{(t,x)=z_0}.$$

Proposition 2.13 (Finite data)**.** *Let Ω be connected and $z_0 \in \Omega$.*

(i) If $\dim_{\mathbb{C}} \mathcal{R}u \leq n$, the jet of order $2n-1$ of u at z_0 determines $\mathrm{Ann}(u)$ among all functions u' with $\dim_{\mathbb{C}} \mathcal{R}u' \leq n$.

(ii) For every u and every M there is a polynomial, hence an exponential-polynomial, with the same jet of order M at z_0; and for finitely many sample points there is an exponential-polynomial with the same values. Consequently, without a bound on $\dim \mathcal{R}u$, no finite set of jets or samples proves $\mathrm{Ann}(u) = (P)$.

Proof. (ii) The Taylor polynomial of order M at z_0 has the same jet, and polynomial interpolation in two variables matches any finite set of values. Both are exponential-polynomials, and an exponential-polynomial never has $\mathrm{Ann} = (P)$ when P has an irreducible factor (Theorem 2.9: all v_i are then exponential-polynomials). (i) The space $\mathcal{R}u$ is invariant under differentiation and, by Lemma 2.7, consists of real-analytic exponential-polynomials. Let K_M be the subspace of elements of $\mathcal{R}u$ whose jet of order M vanishes at z_0. Then $K_{M+1} = \{f \in K_0 : \ \partial_t f, \partial_x f \in K_M\}$, so once $K_{M+1} = K_M$ the chain is constant, and a constant K_M is a derivative-invariant space of functions vanishing at z_0 with all their derivatives, hence $\{0\}$. Since $\dim K_0 \leq n-1$, the chain reaches $\{0\}$ by $M = n-1$: the jet of order $n-1$ at z_0 is injective on $\mathcal{R}u$, and likewise on $\mathcal{R}u'$. For Q of degree at most n, $Q \in \mathrm{Ann}(u)$ iff the jet of order $n-1$ of $Q(\partial)u$

vanishes at z_0, a condition on the jet of order $2n-1$ of u; the same holds for u'. Finally $\mathrm{Ann}(u)$ has codimension at most n, so a Gröbner basis for a degree-compatible order consists of polynomials of degree at most n; the same holds for $\mathrm{Ann}(u')$, and two ideals generated in degree at most n that contain the same polynomials of degree at most n are equal. □

The consequence for learning is a design rule: without an a priori bound on $\dim \mathcal{R}u$, which data never supply, an algorithm can refute a candidate equation, or report that it is compatible with the data at the resolution available, but it cannot certify it. Its output must carry a label that says which of these happened.

2.6 Groups are factors, not terms

Suppose the operator is known and factored as

$$P = P_1^{e_1} \cdots P_N^{e_N}, \qquad Q_j = P_j^{e_j} \text{ pairwise coprime.} \tag{3}$$

Two ways of splitting a solution suggest themselves: one piece per *factor* Q_j, or one piece per additive *term* of the operator (for $u_t = \nu u_{xx} - c u_x$, one piece for diffusion and one for transport). Only the first works.

Theorem 2.14 (Factor decomposition)**.** *Let $\Omega \subset \mathbb{R}^2$ be bounded, connected and simply connected, and let P be as in* (3)*. Every $u \in \ker P(\partial)$ (in $\mathcal{D}'(\Omega)$ or $C^\infty(\Omega)$) can be written*

$$u = \sum_{j=1}^{N} u_j, \qquad Q_j(\partial)\, u_j = 0. \tag{4}$$

The tuples $(u_1, \dots, u_N)$ that sum to zero form a space of exponential-polynomials of dimension exactly

$$\delta(P) = \sum_{i<j} \dim_{\mathbb{C}} \mathcal{R}/(Q_i, Q_j), \tag{5}$$

the number of intersection points of the varieties $V(Q_i)$, counted pair by pair and with multiplicity.

The number $\dim_{\mathbb{C}} \mathcal{R}/(Q_i, Q_j)$ therefore has two readings at once. Geometrically it counts the points where the curves $V(Q_i)$ and $V(Q_j)$ meet, with multiplicity. Analytically it is the dimension of the space of exponential-polynomials that solve both Q_i and Q_j, one exponential for each meeting point (times polynomials at a point of higher multiplicity). For two groups these are exactly the functions that can be moved from one group to the other without changing u: each meeting point of the two curves is one shared mode. For three or more groups $\delta(P)$ still gives the dimension of the ambiguity, but the ambiguous tuples need not be sums of modes shared by pairs.

In one variable the statement is Bézout's identity: if $aQ_1 + bQ_2 = 1$, then $u = bQ_2 u + aQ_1 u$ splits u into a solution of Q_1 and a solution of Q_2. In two variables "coprime" no longer gives $1 \in (Q_1, Q_2)$; it only gives finitely many common zeros, and the finite-dimensional quotient $\mathcal{R}/(Q_1, Q_2)$ measures exactly how much the two groups overlap. The proof combines the exact sequence $0 \to \mathcal{R}/(Q_1 Q_2) \to \mathcal{R}/(Q_1) \oplus \mathcal{R}/(Q_2) \to \mathcal{R}/(Q_1, Q_2) \to 0$ with the vanishing of the obstruction $\mathrm{Ext}^1(M, C^\infty(\Omega))$ for the finite-dimensional module $M = \mathcal{R}/(Q_1, Q_2)$, which reduces to $H^1_{dR}(\Omega) = 0$ (compare the fundamental principle of Ehrenpreis and Palamodov [5, 15]); see Appendix A.

Example 2.15 (Running example: two groups with a two-dimensional overlap). For $Q_1 = \tau + c\xi$ and $Q_2 = \tau - \nu\xi^2$ the common zeros are $(0,0)$ and $(c^2/\nu, -c/\nu)$, both simple, so $\delta(P) = 2$. The ambiguity is spanned by the constant function and by $e^{(c^2/\nu)t-(c/\nu)x}$: a constant (or that growing exponential) can be moved from one group to the other without changing u. Among Fourier modes ($\xi = i\kappa$, $\kappa \in \mathbb{R}$) only the first point is visible: the $\kappa = 0$ mode belongs to both groups.

Projectors from the eigenvalues When each factor is an evolution factor, $Q_j = \tau - \Lambda_j(\xi)$ with pairwise distinct branches Λ_j, a component is an eigenfunction of ∂_t: $\partial_t u_j = \Lambda_j(\partial_x)u_j$. Applying $\prod_{k\neq j}(\partial_t - \Lambda_k(\partial_x))$ to $u = \sum_l u_l$ removes every u_l with $l \neq j$ and multiplies u_j by $\prod_{k\neq j}(\Lambda_j - \Lambda_k)(\partial_x)$. On a spatial Fourier mode κ at which no two branches meet this factor is a nonzero number, hence

$$\hat{u}_j = \prod_{k\neq j} \frac{\partial_t - \Lambda_k(i\kappa)}{\Lambda_j(i\kappa) - \Lambda_k(i\kappa)}\ \hat{u}(t,\kappa). \tag{6}$$

This is a multiplier, rational in κ, with poles exactly where two branches meet ($\Lambda_j(i\kappa) = \Lambda_k(i\kappa)$), that is, at the visible part of the overlap counted by $\delta(P)$. Formula (6) is the Lagrange interpolation formula for the spectral projectors of a matrix with distinct eigenvalues, applied to ∂_t restricted to one wavenumber: the *eigenvalues decide the groups*. For the running example, $\hat{u}_1(t,\kappa) = (\partial_t + \nu\kappa^2)\hat{u}/(\nu\kappa^2 - ic\kappa)$, which is singular only at $\kappa = 0$.

Why additive terms do not form groups

Proposition 2.16 (Terms do not give components)**.** *If $P = \mathcal{A}_1 + \mathcal{A}_2$ with constant coefficients, then* $\ker P \cap (\ker \mathcal{A}_1 + \ker \mathcal{A}_2) \subseteq \ker P \cap \ker(\mathcal{A}_1\mathcal{A}_2)$*, and the right-hand side is finite-dimensional whenever $V(\mathcal{A}_1) \cap V(\mathcal{A}_2)$ is finite.*

Proof. If $u = a + b$ with $\mathcal{A}_1 a = 0$ and $\mathcal{A}_2 b = 0$, then $\mathcal{A}_1\mathcal{A}_2 u = \mathcal{A}_2\mathcal{A}_1 a + \mathcal{A}_1\mathcal{A}_2 b = 0$ because the operators commute. At a common zero of P and $\mathcal{A}_1\mathcal{A}_2$ either $\mathcal{A}_1 = 0$, and then $\mathcal{A}_2 = P - \mathcal{A}_1 = 0$, or $\mathcal{A}_2 = 0$ and then $\mathcal{A}_1 = 0$; so $V(P, \mathcal{A}_1\mathcal{A}_2) = V(\mathcal{A}_1) \cap V(\mathcal{A}_2)$. When this set is finite, the ideal $(P, \mathcal{A}_1\mathcal{A}_2)$ has finite codimension, its common kernel consists of exponential-polynomials with exponents in that set (Lemma 2.7), and it is finite-dimensional. □

For the heat equation, $\mathcal{A}_1 = \partial_t$ and $\mathcal{A}_2 = -\nu\partial_x^2$: every sum $f(x) + a(t) + b(t)x$ of a solution of $\mathcal{A}_1$ and one of $\mathcal{A}_2$ is killed by $\mathcal{A}_1\mathcal{A}_2 \sim \partial_t\partial_x^2$, while a heat mode $e^{-\nu\kappa^2 t}\cos\kappa x$ is not. A generic heat solution is *not* "a diffusion part plus a stationary part". The leftover of any term-wise split is the interaction term of Section 2.8.

Where the decomposition fails The hypotheses of Theorem 2.14 are not technical. On a disconnected domain different pieces can satisfy different equations. On a domain with a hole the decomposition can fail: the Laplacian factors as $(\partial_x + i\partial_y)(\partial_x - i\partial_y)$, yet $\log|z|^2$ is harmonic on an annulus and is not a sum of a holomorphic and an antiholomorphic single-valued function. The obstruction is the first de Rham cohomology of Ω, which is why the algorithm works on rectangles.

Role in the algorithm. Theorem 2.14 is the grouping principle of the paper. The objects that a learning algorithm should group are *modes of the same factor*; equivalently, points of the data's Fourier–Laplace spectrum lying on the same irreducible component of $V(P)$. The

overlap $\delta(P)$ is the irreducible amount of ambiguity, and the algorithm resolves it by a fixed convention (Section 4.6).

2.7 How well separated are the groups? The Friedrichs angle

Theorem 2.14 is algebraic: it says that the split exists, not that it is stable. Stability is a geometric property of the component spaces $M_j = \ker_{H^s} Q_j$, the solutions of each factor in the Sobolev space $H^s(\Omega)$. Let

$$\mathcal{S}:\ \textstyle\bigoplus_{j=1}^{N} M_j \to H^s(\Omega), \qquad \mathcal{S}(u_1,\dots,u_N) = \sum_j u_j,$$

be the *sum operator*, with $\|(u_1,\dots,u_N)\|^2 = \sum_j \|u_j\|_{H^s}^2$, and define the *conditioning constant*

$$\gamma_N = \inf_{0\neq v \perp \ker \mathcal{S}} \frac{\|\mathcal{S}v\|^2}{\|v\|^2} = \inf \sigma\big(\mathcal{S}^*\mathcal{S}\big|_{(\ker \mathcal{S})^\perp}\big). \tag{7}$$

It is the worst case, over all tuples of pieces, of the size of the sum relative to the size of the pieces. Tuples in $\ker \mathcal{S}$ (the overlap of Theorem 2.14) are excluded. By the triangle and Cauchy–Schwarz inequalities, $\|\sum_j u_j\|^2 \le \big(\sum_j \|u_j\|\big)^2 \le N \sum_j \|u_j\|^2$, hence

$$0 \le \gamma_N \le N.$$

If the spaces are mutually orthogonal, $\|\sum_j u_j\|^2 = \sum_j \|u_j\|^2$ and $\gamma_N = 1$. At the other end, if M_2 approaches M_1 one can take $u_2 \approx -u_1$: the pieces stay of unit size while their sum becomes arbitrarily small, so $\gamma_N \to 0$. A small γ_N is the signature of groups that can almost cancel each other.

For two groups, $\gamma_2 = 1 - c_F$, where the *Friedrichs cosine*

$$c_F = \sup\big\{|\langle a, b\rangle| :\ a \in M_1 \ominus (M_1 \cap M_2),\ b \in M_2 \ominus (M_1 \cap M_2),\ \|a\| = \|b\| = 1\big\} = \cos\theta_F$$

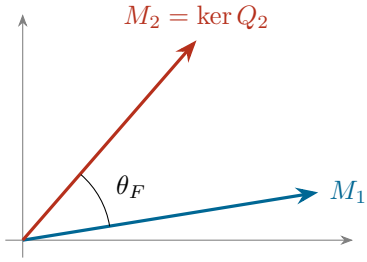
is the cosine of the smallest nonzero angle θ_F between the two spaces (Fig. 3) [4]. In regression terms, M_1 and M_2 are two blocks of regressors and c_F is the largest correlation between a combination of the first block and one of the second: the closer the blocks are to collinear, the less the data can attribute to each.

Theorem 2.17 (Stability of the split)**.** *Every $u \in \ker_{H^s} P$ has components $u_j \in H^s$ if and only if $\sum_j M_j$ is closed and dense in $\ker_{H^s} P$. Closedness is equivalent to $\gamma_N > 0$, and then*

$$\Big(\sum_j \|u_j\|_{H^s}^2\Big)^{1/2} \le \gamma_N^{-1/2}\, \|u\|_{H^s} \tag{8}$$

for the minimum-norm components. If the component spaces are finite-dimensional, as for the sampled kernels used by the algorithm, the sum is always closed and density holds whenever the sampled kernel of P is spanned by those of its factors, so stability reduces to $\gamma_N > 0$.

Density in H^s is a genuine condition on infinite-dimensional spaces. On convex domains solutions of $P(\partial)u = 0$ are limits of exponential solutions, uniformly on compact subsets [12, 9], and every exponential solution lies in $\sum_j M_j$; whether this extends to the H^s norm up to the boundary for general P we do not claim.



$c_F = \cos\theta_F, \quad \gamma_2 = 1 - c_F$
error amplification $\gamma_2^{-1/2}$
running example: $\theta_F \approx 0.60$, $\gamma_2 \approx 0.17$

Figure 3: The Friedrichs angle θ_F between the solution spaces of two factors. A small angle means that large pieces can almost cancel, so recovering the pieces from their sum amplifies errors by up to $\gamma_2^{-1/2} = (1-\cos\theta_F)^{-1/2}$. The values quoted are those measured on the running example in Section 6.1.

Two warnings come with the theorem. First, $\gamma_N > 0$ can fail on infinite-dimensional component spaces: the pieces of a smooth u may then be one derivative rougher than u itself. Second, for $N \geq 3$ groups the pairwise cosines do not determine γ_N: two pairs can be well separated while the three spaces are nearly dependent, so the full Gram operator of $\mathcal{S}$ must be used.

Role in the algorithm. γ_N is computed on the sampled kernel bases and reported with every result. It enters twice: as the error amplification $\gamma_N^{-1/2}$ of the components, and through the Bayesian split layer (Section 5.1), whose credible intervals widen and whose split probability returns to the prior as $\theta_F \to 0$. A small angle is therefore never hidden: it shows up as wide intervals, not as a confident but wrong split.

2.8 Inside one group: the interaction term and its energy

A single factor may have several additive terms: $Q_2 = \tau - \nu\xi^2$ has one, but the advection–diffusion factor $\tau + c\xi - \nu\xi^2$ has two. Proposition 2.16 says that such a factor cannot be split further into one solution per term. We now measure exactly what is left over.

Consider an evolution $u_t = \mathcal{A}u$ with $\mathcal{A} = \mathcal{A}_1 + \cdots + \mathcal{A}_K$ (the terms), the flow $\Phi(t) = e^{t\mathcal{A}}$ and the flows of the single terms $\Phi_n(t) = e^{t\mathcal{A}_n}$. If term n acted alone it would change the datum by $\Delta_n u_0 = (\Phi_n(t) - \mathrm{Id})u_0$. The *interaction term* is the part of the evolution that is not the sum of these single-term changes,

$$E_{\mathrm{int}}(t) = u(t) - u_0 - \sum_{n=1}^{K} (\Phi_n(t)u_0 - u_0). \tag{9}$$

Theorem 2.18 (Interaction term)**.** *If the terms commute (in particular, for constant coefficients),*

$$E_{\mathrm{int}}(t) = \Big[\prod_{n=1}^{K}(\mathrm{Id} + \Delta_n) - \mathrm{Id} - \sum_{n=1}^{K}\Delta_n\Big]u_0 = \sum_{\mathcal{E}\subseteq\{1,\dots,K\},\ |\mathcal{E}|\geq 2} \Big(\prod_{n\in\mathcal{E}}\Delta_n\Big)u_0.$$

For two terms $E_{\mathrm{int}} = \Delta_1\Delta_2 u_0$*. On Fourier modes, with symbols* $\hat{\mathcal{A}}_n(\kappa)$ *of the terms,*

$$\widehat{E_{\mathrm{int}}}(t,\kappa) = \Big[\prod_n e^{t\hat{\mathcal{A}}_n(\kappa)} - 1 - \sum_n (e^{t\hat{\mathcal{A}}_n(\kappa)} - 1)\Big]\hat{u}_0(\kappa). \tag{10}$$

Proof. Commuting terms give $\Phi = \prod_n \Phi_n = \prod_n (\mathrm{Id} + \Delta_n)$. Expanding the product over subsets $\mathcal{E} \subseteq \{1, \dots, K\}$ and removing the subsets with zero or one element gives the formula; on Fourier modes every operator is multiplication by a number. □

Formula (10) costs $O(K)$ operations per wavenumber, although writing out all cross terms would need $2^K - K - 1$ of them. For advection–diffusion, $|\widehat{E_{\mathrm{int}}}|/|\hat{u}_0| = |(e^{-ic\kappa t} - 1)(e^{-\nu\kappa^2 t} - 1)| \leq 2$: the interaction is of the same size as the solution, so it is not a small correction that a regularizer could remove.

Proposition 2.19 (No stable removal)**.** *Let $Q = \tau + c\xi - \nu\xi^2$ (advection–diffusion, terms $-c\partial_x$ and $\nu\partial_x^2$) on a bounded rectangle $\Omega = (0, T) \times (a, b)$. For every $s \geq 0$,*

$$\sup \left\{ \|E_{\mathrm{int}}\|_{L^2(\Omega)} / \|u\|_{H^s(\Omega)} : \ u \in \mathrm{EP}, \ Q(\partial)u = 0, \ u \neq 0 \right\} = \infty,$$

so $u \mapsto E_{\mathrm{int}}$ has no continuous extension to $\ker_{H^s} Q$. Moreover, a solution of $Q(\partial)u = 0$ is a sum $v + w$ of solutions of the single-term evolutions $(\partial_t + c\partial_x)v = 0$ and $(\partial_t - \nu\partial_x^2)w = 0$ only if it lies in a finite-dimensional space of exponential-polynomials.

The test solutions are $e^{\lambda_\zeta t + \zeta x}$ with real $\zeta > c/\nu$ and $\lambda_\zeta = \nu\zeta^2 - c\zeta$, legitimate data on a bounded interval: diffusion alone amplifies them by $e^{\nu\zeta^2 t}$, the solution grows only like $e^{\lambda_\zeta t}$, and the interaction must cancel a difference that grows faster than any power of ζ (Appendix A). On a periodic domain with bounded wavenumbers the Fourier multiplier above is bounded by 2; the instability comes from the non-periodic window, which is the situation of real data. The second statement means that a model which writes the data as a sum of single-term pieces, with any penalty on the pieces, leaves a remainder on generic data, and that remainder is the interaction.

Energy We measure the size of a group and of the interaction by L^2 energies on the observation window. The *energy share* of group j and the *interaction ratio* are

$$w_j = \frac{\|u_j\|}{\sum_k \|u_k\|}, \qquad \rho = \frac{\|E_{\mathrm{int}}(T)\|}{\|u(T) - u_0\|}, \tag{11}$$

where T is the length of the time window. The shares w_j say how much of the data each group explains; ρ says how far the evolution inside a multi-term factor is from the superposition of its terms ($\rho = 0$ for a factor with a single term). Interactions occur only *inside* a factor: different factors superpose exactly, which is another way of saying that factors, not terms, are the right groups.

Role in the algorithm. By Proposition 2.19, a value of E_{int} computed from one interpolated field on a bounded window is not reliable. The algorithm reports the distribution of ρ over posterior samples (Section 5.2), never a single number, and uses the shares w_j with credible intervals.

3 Nonlinear equations: why grouping gets harder

This section is an outlook: it explains which parts of the grouping structure are lost for nonlinear equations and which survive, and a reader interested in the method can go directly

to Section 4. The one fact used later is that the interaction term, unlike the groups, is still defined.

The results above rest on one structure: candidate equations form the commutative ring $\mathcal{R} = \mathbb{C}[\tau, \xi]$, solutions form vector spaces, and the exact sequence behind Theorem 2.14 (a Chinese remainder theorem up to the finite overlap $\delta(P)$) turns a factorization of the symbol into a splitting of the solution. For a nonlinear equation $u_t = \sum_n \mathcal{A}_n(u)$, with each $\mathcal{A}_n$ a polynomial in the jet $(u, u_x, u_{xx}, \dots)$, this structure is lost in three ways.

(i) *No ring of candidates.* Products of nonlinear operators are not operators of the same kind in any useful sense, and there is no unique factorization. One fixes instead a finite *dictionary* $\mathcal{D}$ of jet monomials of bounded degree and order, with span $\mathcal{V}$, and asks which relations $F(u, u_x, \dots) = 0$ in $\mathcal{V}$ the data satisfy. In practice each monomial (for instance u_t, u_x, u_{xx}, uu_x, u^2) is evaluated at the sample points, giving a matrix Θ with one column per monomial; a relation $F = \sum_m c_m\, m(u)$ is a vector c with $\Theta c \approx 0$, so the candidate equations are the near-null space of Θ.

(ii) *No superposition.* Solutions do not form a vector space, so "the component of u along a factor" has no meaning; the groups of Theorem 2.14 do not exist.

(iii) *Identification is relative.* Identification can only hold relative to $\mathcal{V}$ and modulo the differential consequences of F (its derivatives $\partial_x F, \partial_x^2 F, \dots$ are also relations in any rich dictionary). The precise statement is an exact sequence relating the relations of u in $\mathcal{V}$ to those generated by F; identification holds exactly when a jet-level obstruction vanishes.

What survives is the interaction term. Definition (9) still makes sense with the nonlinear single-term flows, and its short-time expansion is

$$E_{\mathrm{int}}(t) = \frac{t^2}{2} \sum_{n \neq m} \mathcal{A}_n'[u_0](\mathcal{A}_m(u_0)) + O(t^3), \tag{12}$$

where $\mathcal{A}_n'[u_0]$ is the Fréchet derivative of $\mathcal{A}_n$ at u_0. For Burgers' equation $u_t = -uu_x + \nu u_{xx}$ the two terms are $\mathcal{A}_1(u) = -uu_x$ and $\mathcal{A}_2(u) = \nu u_{xx}$, with Fréchet derivatives $\mathcal{A}_1'[u](v) = -(vu_x + uv_x)$ and $\mathcal{A}_2'[u](v) = \nu v_{xx}$. Then

$$\mathcal{A}_1'[u_0](\mathcal{A}_2(u_0)) = -\nu\,(u_0'' u_0' + u_0 u_0'''), \qquad \mathcal{A}_2'[u_0](\mathcal{A}_1(u_0)) = -\nu\,(3u_0' u_0'' + u_0 u_0'''),$$

and half their sum gives

$$E_{\mathrm{int}}(t) = -\nu\,(u_0 u_0''' + 2u_0' u_0'')\, t^2 + O(t^3).$$

The expansion holds only while the solution is smooth. The flow of the single term $-uu_x$ is the inviscid Burgers equation, whose characteristics cross: for $u_0 = \sin x$ they are $x = \xi + t \sin \xi$, and $\partial x / \partial \xi = 1 + t \cos \xi$ first vanishes at $\xi = \pi$, $t^* = 1$. After this breaking time the single-term flow has a jump, E_{int} has a jump in x whose height does not vanish as $\nu \to 0$, and the interaction can only be measured in L^1, where for a single shock it is small, of order ν up to logarithmic factors (a rate we do not prove here). Whether such data still identify the equation relative to a dictionary depends on how many shocks they contain compared with the degree of the dictionary; we do not pursue this here.

Variable coefficients If the coefficients depend on (t, x), the operators form the Weyl algebra $\mathbb{C}[t, x]\langle \partial_t, \partial_x \rangle$, which is not commutative ($\partial_x\, x - x\, \partial_x = 1$). Annihilators become left ideals,

Table 3: What carries over beyond scalar linear equations with constant coefficients. "Groups" means a splitting of a solution into pieces attached to pieces of the equation.

Setting	Candidate equations	Groups	What survives
linear, constant coefficients	commutative ring $\mathbb{C}[\tau, \xi]$; ideals; unique factorization	factors of the symbol, overlap $\delta(P)$ (Theorem 2.14)	everything in this paper
variable coefficients	non-commutative Weyl algebra; left ideals; factorization not unique	no general splitting theorem	identification inside a finite-dimensional class of coefficients; local rates $\lambda(\kappa; x)$ (Remark 4.2)
systems (e.g. Dirac)	matrices over the ring; modules instead of ideals	sums of kernels need extra conditions and can fail on non-convex domains	identification criteria for square non-singular systems
nonlinear	finite dictionary of jet monomials; differential ideal	none (no superposition)	identification relative to the dictionary; the interaction term and its expansion (12)

factorizations are no longer unique, and Theorem 2.14 has no general analogue. Worse, if the coefficients may be arbitrary smooth functions, one solution identifies nothing: any u without zeros satisfies the first-order equation $(\partial_x - u_x/u)u = 0$, whose coefficient is built from u itself. Identification becomes possible only inside a fixed, finite-dimensional class of coefficients. For the Schrödinger equation $iu_t = -u_{xx} + V(x)u$, for instance, the potential is read off directly as $V(x) = (iu_t + u_{xx})/u$ wherever $u \neq 0$. In the spectral picture of Section 4.4, slowly varying coefficients would make the rates depend on position, $\lambda(\kappa; x)$ (Remark 4.2).

A possible route The grouping idea can be carried over by replacing "modes on the same irreducible component of $V(P)$" with "jet samples satisfying the same relation in $\mathcal{V}$". Locally in time, (12) shows that a nonlinear flow is a linear one (its Fréchet derivative) plus a quadratic correction, so one can group linearized modes on short windows and test, through the size of E_{int}, whether the groups persist. We do not pursue this here: without a ring there is no analogue of $\delta(P)$, of the projectors (6) or of γ_N, and these are what make the linear algorithm verifiable.

4 The algorithm

The algorithm turns the structure of Section 2 into a procedure on data. It interpolates the observations into field samples (Section 4.1) and restricts attention to the wavenumbers that carry information (Section 4.2); it runs an outer loop over classes of operators (Section 4.3); inside the loop it reads the eigenvalues of each wavenumber with a matrix pencil (Section 4.4), groups them into factors (Section 4.5), recovers the groups as functions (Section 4.6), measures how much they interfere and recurses on each of them (Section 4.7), and labels the result (Section 4.8). Algorithm 1 in Appendix B lists all steps.

4.1 From scattered observations to field samples

The input is a set of noisy observations $y_i = u(t_i, x_i) + \varepsilon_i$, $i = 1, \dots, n$, with $\varepsilon_i \sim \mathcal{N}(0, \sigma_n^2)$, on a window $[0, T] \times [0, L)$ periodic in x, or a noisy field on a regular $n_t \times n_x$ grid. Every later stage works on Fourier modes, so it needs fields on a grid.

Gaussian-process prior on the field We model $y = \bar{y} + f + \varepsilon$ with a constant $\bar{y}$ and

$$f \sim \mathcal{GP}\Big(0,\ \sum_{r=1}^{R} \eta_r^2\, k_t^{(r)}(t, t')\, k_x^{(r)}(x, x')\Big), \tag{13}$$

a sum of R separable Matérn-5/2 kernels with their own amplitudes η_r and length scales $\ell_t^{(r)}, \ell_x^{(r)}$ (k_x periodic in x). More than one product is used because the groups we look for have different correlation lengths: a transported profile and a diffusing one do not share a scale. The hyperparameters $\eta = (\eta_r, \ell^{(r)}, \sigma_n, \bar{y})$ are not fixed at their maximum; they are integrated out,

$$p(f \mid y) = \int p(f \mid y, \eta)\, p(\eta \mid y)\, \mathrm{d}\eta \ \approx\ \sum_{h=1}^{N_\eta} \pi_h\, p(f \mid y, \eta_h), \qquad \pi_h \propto \frac{p(y, \eta_h)}{q(\eta_h)}, \tag{14}$$

where the η_h are drawn from a Laplace (Gaussian) approximation q of $p(\eta \mid y)$ and the importance weights π_h correct for the difference between q and the true posterior. Each $p(f \mid y, \eta_h)$ is an ordinary Gaussian process posterior, so the result is a weighted mixture of Gaussian processes.

Samples, not the mean The output of this stage is N_s fields $u^{(q)}$ on the grid, drawn from the posterior by pathwise (Matheron) conditioning: each sample is drawn from one of the Gaussian processes $p(f \mid y, \eta_h)$ of (14) and carries its weight, $\pi_q = \pi_h$ normalized over the samples. We do not use the posterior mean. The maps applied next are not continuous: the interaction term is not (Proposition 2.19), and the factors of a numerical symbol jump where it becomes reducible (Section 4.5), so the mean of the outputs differs from the output of the mean. In addition the mean is a Wiener filter: on a Fourier mode with prior spectral density $S(\kappa)$ it multiplies the data by $S(\kappa)/(S(\kappa) + \sigma_n^2/n_\mathrm{eff})$, with n_eff the effective number of observations per mode, which shrinks fast-decaying, high-κ modes towards zero. Samples keep the posterior spread; every downstream quantity is reported as a distribution over them. On gridded input the noisy grid itself is used, with parametric-bootstrap refits providing the spread.

4.2 Nyquist limits and the high-frequency band

Which wavenumbers carry usable information about the symbol? Three limits apply, and the algorithm uses all three to choose the band $\mathcal{K}$ of wavenumbers it analyses.

Spatial aliasing On a grid of spacing Δx the discrete Fourier transform sees wavenumbers $|\kappa| \le \kappa_x^{\max} = \pi/\Delta x$; a mode with larger $|\kappa|$ appears at $\kappa - 2\pi q/\Delta x$ for an integer q. An aliased mode keeps its temporal rate $\lambda(\kappa)$ but is placed at the wrong wavenumber, so it produces a point $(\lambda, \kappa - 2\pi q/\Delta x)$ that is *not* on the variety. For scattered data there is no strict Nyquist limit, but the mean spacing $h = \sqrt{TL/n}$ plays the same role.

Temporal aliasing along a branch Rates are read from a pencil of matrices built from samples spaced by Δt, and log of an eigenvalue returns $\operatorname{Im}\lambda$ only modulo $2\pi/\Delta t$. A branch $\tau = \Lambda_j(\xi)$ is read without ambiguity on the wavenumbers where

$$|\operatorname{Im}\Lambda_j(i\kappa)| < \pi/\Delta t. \tag{15}$$

For transport at speed c this means $|\kappa| < \pi/(c\,\Delta t)$: the faster the transport, the narrower the band.

Noise floor and decay The observation error is modelled as Gaussian white noise with standard deviation σ_n per grid point. After the normalized Fourier transform in x, one wavenumber bin carries noise of standard deviation $\sigma_\kappa = \sigma_n/\sqrt{n_x}$. A mode is counted as detected while its amplitude exceeds z times this bin noise; $z\,\sigma_\kappa$ is the *detection threshold* ($z = 3$ by default). A mode with initial amplitude A_κ and rate $\lambda(\kappa)$ has amplitude $|A_\kappa|e^{\operatorname{Re}\lambda(\kappa)t}$ at time t, so it reaches the threshold when

$$|A_\kappa|\,e^{\operatorname{Re}\lambda(\kappa)\,t} = z\,\sigma_\kappa, \qquad \text{that is} \qquad t = \frac{\log\left(|A_\kappa|/(z\,\sigma_\kappa)\right)}{-\operatorname{Re}\lambda(\kappa)}.$$

Capping by the window length and treating non-decaying modes and modes that start below the threshold gives the time during which the mode is visible,

$$T_{\mathrm{vis}}(\kappa) = \min\Big\{T,\ \frac{\max\left(0, \log(|A_\kappa|/(z\,\sigma_\kappa))\right)}{\max(0, -\operatorname{Re}\lambda(\kappa))}\Big\}. \tag{16}$$

A rate can be estimated only if $T_{\mathrm{vis}}(\kappa)$ covers several samples; for a diffusive branch $\operatorname{Re}\lambda = -\nu\kappa^2$, so the usable band ends near $\kappa \approx \left(\log(|A_\kappa|/z\sigma_\kappa)/(\nu\, n_{\min}\Delta t)\right)^{1/2}$, where $n_{\min}$ is the smallest number of time samples the pencil needs. High wavenumbers of a dissipative group are therefore lost first. This is a property of the data, not of the method; interpolation can only make it worse, by the shrinkage described above. Figure 4 shows the three limits on the running example.

Wavenumbers per group The band must contain enough distinct wavenumbers on each branch for the factor to be determined.

Proposition 4.1 (Counting condition)**.** *Let $Q = \tau - \Lambda(\xi)$ with Λ a polynomial of degree at most d with complex coefficients. The coefficients are determined by the rates $\lambda(\kappa_\ell) = \Lambda(i\kappa_\ell)$ at $d+1$ distinct wavenumbers $\kappa_\ell \in \mathbb{R}$, and not by fewer. The coefficient error is bounded by $\|\mathsf{V}^{-1}\|\max_\ell|\delta\lambda_\ell|$, where $\mathsf{V} = \left((i\kappa_\ell)^k\right)$ is the Vandermonde matrix of the nodes; $\|\mathsf{V}^{-1}\|$ grows like h_κ^{-d} when the nodes cluster, where $h_\kappa = \max_{\ell,\ell'}|\kappa_\ell - \kappa_{\ell'}|$ is their spread. If the coefficients are real, the rates at $-\kappa$ are the conjugates of those at κ, and $\lfloor d/2\rfloor + 1$ distinct values of $|\kappa|$ suffice, at least $\lceil d/2\rceil$ of them nonzero.*

Proof. $\Lambda(i\kappa) = \sum_{k\le d}\beta_k(i\kappa)^k$ is linear in $(\beta_0, \dots, \beta_d)$ with matrix V, which is invertible iff the $d+1$ nodes $i\kappa_\ell$ are distinct; with d nodes a polynomial of degree d vanishing at all of them can be added. The bound is the norm of the inverse; for nodes within h_κ of each other the Lagrange basis polynomials have coefficients of size h_κ^{-d} [7]. For real β_k, $\operatorname{Re}\Lambda(i\kappa) = \sum_k(-1)^k\beta_{2k}\kappa^{2k}$ and $\operatorname{Im}\Lambda(i\kappa) = \sum_k(-1)^k\beta_{2k+1}\kappa^{2k+1}$ separate the even and the odd coefficients, which are polynomials in κ^2 with $\lfloor d/2\rfloor + 1$ and $\lceil d/2\rceil$ unknowns. □

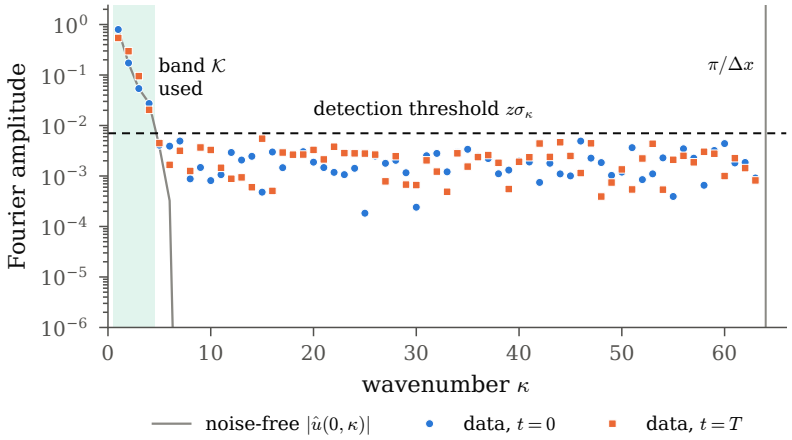


Figure 4: Fourier amplitudes of the worked example of Section 6.1 (running example, grid, 5% noise) at the first and last time, against the per-bin detection threshold $z\,\sigma_\kappa$ ($z = 3$). Only the shaded band of wavenumbers carries information about the symbol; the spatial Nyquist limit $\pi/\Delta x$ is far away and is not the binding constraint. What limits the band is the noise floor together with the decay of the diffusive modes (Section 4.2).

Role in the algorithm. The band $\mathcal{K}$ is the set of wavenumbers below both Nyquist limits whose amplitude exceeds $z\,\sigma_\kappa$ (default $z = 3$). A factor of degree d_j in ξ is reported as FIT-COMPATIBLE only if its modes occupy at least $d_j + 1$ distinct signed wavenumbers $\kappa \in \mathcal{K}$ (for a real field the modes at $\pm\kappa$ come in conjugate pairs, and the rule then implies the real-coefficient count of the proposition), and the whole band is used rather than the cleanest few bins, because spreading the nodes is what keeps V well conditioned. Figure 4 shows the band for the running example.

4.3 The outer loop: choosing the class of operators

A *structure* $\mathsf{M} = ((m_1, d_1), \ldots, (m_N, d_N))$ fixes the number N of factors and, for each, its degree m_j in τ and d_j in ξ; a factor is monic in τ. For the running example the true structure is $\mathsf{M} = ((1,1),(1,2))$: one factor of degree one in τ and one in ξ ($\tau + c\xi$), and one of degree one in τ and two in ξ ($\tau - \nu\xi^2$). The *class* $\mathcal{C}(m,d)$ contains all structures with $\sum_j m_j \le m$ and every $d_j \le d$. The starting class $\mathcal{C}(3,4)$ contains $((1,1),(1,2))$, but also, for instance, $((2,2))$ (a single second-order factor, such as the wave operator $\tau^2 - c^2\xi^2$) and $((1,0),(1,1),(1,1))$ (three first-order factors); it has 90 structures, and the largest class allowed, $\mathcal{C}(4,8)$, has 1353. For each structure the core (Sections 4.4 and 4.5) returns fitted factors and a fit score.

Comparing structures The structures are compared by the Bayesian information criterion

$$\mathrm{BIC}(\mathsf{M}) = 2n_{\mathrm{m}} \log \max\left(\chi^2/(2n_{\mathrm{m}}),\, 1\right) + n_{\mathrm{par}}(\mathsf{M}) \log(2n_{\mathrm{m}}), \tag{17}$$

where n_{m} is the number of fitted modes (each complex rate gives two real data, hence $2n_{\mathrm{m}}$), $n_{\mathrm{par}}(\mathsf{M})$ is the number of free coefficients of the structure, and

$$\chi^2 = \sum_{p=1}^{n_{\mathrm{m}}} \min\left(r_{j(p)}(p)^2,\, 9\right)$$

measures how badly the fitted factors explain the observed modes, through the distance $r_j(p)$ of each mode to the factor it is assigned to, in units of its standard deviation ((21) in Section 4.5). If the structure is right and the rate errors are circular complex Gaussian, each $r_j(p)^2$ is the squared modulus of a standard complex Gaussian, exponentially distributed with mean one, so that, ignoring the cap, χ^2 follows over noise realizations approximately one half of a chi-squared distribution with $2n_{\mathrm{m}} - n_{\mathrm{par}}$ degrees of freedom, and $\chi^2/(2n_{\mathrm{m}})$ is close to $\frac{1}{2}$. The first term of (17) penalizes a poor fit and the second penalizes complexity; because of the floor $\max(\cdot, 1)$, misfit is penalized only once it clearly exceeds what the noise explains, and the criterion never rewards a fit that is better than the noise level, which would only be fitting noise. The criterion is turned into a probability over structures,

$$P(\mathsf{M} \mid \text{data}) \propto e^{-\mathrm{BIC}(\mathsf{M})/2}, \tag{18}$$

so that every structure receives a score that trades its fit against its number of coefficients.

Growing the class The class is grown rather than fixed in advance. Starting from $(m, d) = (3, 4)$, the algorithm raises m or d by one when (a) most of the posterior mass (18) is on structures labelled MISFIT or ID-KER (Section 4.8), or (b) an omnibus goodness-of-fit test rejects the current fit, $\log_{10} p < -3$, where p is the p-value of the test under the hypothesis that the fitted structure explains the data. A larger class almost always fits better, simply because it has more degrees of freedom, so it is kept only if two further checks pass. The *held-out check* fits the modes of the earlier part of the time window and predicts those of the later part: a structure that captures the dynamics predicts data it was not fitted to, while one that only improves the fit to the data it has seen is overfitting. The *mass-shift check* asks that the larger class move posterior mass towards structures labelled FIT-COMPATIBLE. Growth stops at $(4, 8)$.

Regrouping The loop runs on every field sample $u^{(1)}, \ldots, u^{(N_{\mathrm{s}})}$, and different samples may give slightly different answers: two factors in one sample and three in another, say. Coefficients of such answers cannot be compared directly. The algorithm therefore takes the consensus structure, the one with the largest posterior mass over all samples, and refits every sample at that structure (*regrouping*); after this step every sample contributes one fitted factor of each type, and factors from different samples can be compared and clustered (Section 5.2).

This loop is the only "regularization" in the method, and it is a regularization of the *class*, by Occam's razor, not a penalty on coefficients. A Tikhonov or sparsity penalty would be the wrong tool: it cannot make the interaction vanish (Proposition 2.19), and it can push the error of the split below the unbiased floor σ_n^2/γ_N (Section 7) only by shrinking the components towards a prior, which the split layer of Section 5.1 does explicitly and reports.

4.4 Reading the eigenvalues: the matrix pencil

For each sample field the spatial Fourier transform gives, for every $\kappa \in \mathcal{K}$, a time series $\hat{u}(t_\ell, \kappa)$, $\ell = 0, \ldots, n_t - 1$. For a constant-coefficient equation it is an exponential sum,

$$\hat{u}(t, \kappa) = \sum_{p=1}^{r(\kappa)} A_p \, e^{\lambda_p(\kappa) t}, \qquad Q_{j(p)}(\lambda_p, i\kappa) = 0, \tag{19}$$

one term per root of the symbol at that wavenumber. This step does *not* fit the factors of the equation. It extracts, at each wavenumber separately, the temporal rates $\lambda_p(\kappa)$, which are the eigenvalues of ∂_t restricted to the finite-dimensional space spanned by the time series and its shifts, exactly as in Lemma 2.7. Because every rate satisfies $Q_j(\lambda_p, i\kappa) = 0$ for one factor, the factors are recovered afterwards by finding the relation between λ and κ, that is, the dispersion relation of each group (Section 4.5): the search for the equation happens in frequency.

We compute the rates with the matrix pencil method [10]. From the time series at one wavenumber form the Hankel matrix $H_{\ell k} = \hat{u}(t_{\ell+k}, \kappa)$, whose columns are copies of the time series shifted by one step each, and estimate the number of exponentials r from its singular values. Let Y_0 be H without its last column and Y_1 be H without its first column, both projected on the r leading singular directions. Advancing a time series by one step multiplies its term $A_p e^{\lambda_p t}$ by $\mu_p = e^{\lambda_p \Delta t}$, so Y_1 is Y_0 with each exponential multiplied by its own μ_p, and these factors are the eigenvalues of the pencil:

$$\lambda_p = \frac{1}{\Delta t} \log \mu_p, \qquad \mu_p \in \operatorname{eig}\left(Y_0^{+} Y_1\right), \tag{20}$$

followed by a least-squares fit of the amplitudes A_p. A bin is used only if its amplitude exceeds $z\,\sigma_\kappa$, and the order r is selected per bin by BIC among r_0, r_0+1, r_0+2. The uncertainty of each rate is taken from the Cramér–Rao bound, the smallest variance that any unbiased estimator of λ_p can reach at this noise level. It depends on how sensitive the time series is to a change of each parameter: with J the Jacobian of (19) with respect to all rates and amplitudes in the bin, $\sigma^2_{\lambda,p} = \sigma^2_\kappa [(\mathsf{J}^H \mathsf{J})^{-1}]_{pp}$. Rates of weak or quickly decaying modes, or of two modes with nearly equal rates, are sensitive to noise and receive a large $\sigma_{\lambda,p}$.

The output is a cloud of points $(\kappa, \lambda_p(\kappa))$ with error bars. The modes of one group lie on one branch of the dispersion relation; Fig. 5 shows these branches for the running example. Two rates closer than about $4/T$ cannot be resolved in a window of length T; such a bin is flagged as *confluent* and the factors involved are labelled UNDECIDED rather than forced apart.

Remark 4.2 (Variable coefficients and higher dimension). For constant coefficients the rates $\lambda_p(\kappa)$ do not depend on x; this is what allows one spectral decomposition per wavenumber. If the coefficients vary slowly in x, a windowed transform gives local rates $\lambda_p(\kappa; x_0)$, and each group becomes a *family* of branches indexed by x_0; the structured fit below would then regress coefficient functions instead of numbers. In d space dimensions the same construction applies with a wavenumber vector $\kappa \in \mathbb{R}^d$: the pencil still reads the temporal rates at each κ, one small eigenvalue problem per wavenumber, and the groups become hypersurfaces $Q_j(\lambda, i\kappa) = 0$ in $\mathbb{C} \times \mathbb{C}^d$. This makes the pencil a natural, though not necessarily the most efficient, first layer for decomposing equations in higher dimension. What is missing is the second layer: a decomposition theorem for symbols in $d+1$ variables, where two factors can share a whole curve of zeros and the overlap is no longer finite-dimensional. Neither extension is implemented, and the algebraic guarantees of Section 2.6 do not carry over unchanged (for variable coefficients the ring becomes non-commutative).

4.5 Grouping the modes: the structured fit

The natural route, fitting one polynomial P through all points $(\lambda_p, i\kappa_p)$ and then factoring it, is ill-posed: a numerical polynomial with noisy coefficients is almost never reducible, and

when it is the factors move discontinuously with the coefficients. In our tests this route split the irreducible Klein–Gordon and telegraph symbols into spurious factors at noise 10^{-4}.

We group and fit at the same time. For a structure M with N factors we solve the N-*varieties* problem, a version of k-means clustering in which the cluster centres are curves instead of points: assign every mode to the factor whose variety is nearest, refit each factor to the modes assigned to it, and repeat the two steps until the assignment no longer changes (8 random starts, 30 iterations). The distance of a mode to factor Q_j is the first-order distance to its variety, scaled by the rate uncertainty,

$$r_j(p) = \frac{|Q_j(\lambda_p, i\kappa_p)|}{\sigma_{\lambda,p}\,|\partial_\tau Q_j(\lambda_p, i\kappa_p)|}, \tag{21}$$

which plays the role of the Euclidean distance in k-means, and the misfit of a mode is $\min(r_j(p)^2, 9)$; the cap makes isolated outliers count as a fixed penalty instead of dragging a factor away. This is a clustering problem in which the clusters are algebraic curves rather than points: the grouping of Theorem 2.14 learned directly from the spectrum.

4.6 Recovering the groups as functions

Once modes are assigned, the component of group j is the sum of its modes,

$$u_j(t,x) = \sum_{p:\, j(p)=j} A_p\, e^{\lambda_p(t-t_0)+i\kappa_p x}, \tag{22}$$

the discrete counterpart of the projector (6). A mode within distance 3 of two varieties lies in the overlap counted by $\delta(P)$ (for the running example, the $\kappa = 0$ mode); it is split equally between the groups. This is the minimum-norm, or minimum-energy, choice among all splits allowed by Theorem 2.14: if the shared mode has amplitude A and is given as αA to one group and $(1-\alpha)A$ to the other, the energy it adds to the components is $|A|^2(\alpha^2 + (1-\alpha)^2)$, which is smallest at $\alpha = \frac{1}{2}$. The data cannot tell the splits apart, so the algorithm takes the one that puts the least energy into the ambiguity. It coincides with the posterior mean of the Bayesian layer (Section 5.1), so the convention is not arbitrary. The energy shares w_j of (11) are computed from these components.

4.7 Interference between groups and the recursion over components

The groups are not orthogonal. Let B_j be an orthonormal basis of the sampled kernel of Q_j (all exponentials $e^{\lambda t + i\kappa x}$ with $Q_j(\lambda, i\kappa) = 0$, $\kappa \in \mathcal{K}$, on the grid) and $\mathcal{S} = [B_1 \cdots B_N]$. Then γ_N is the smallest eigenvalue of $\mathcal{S}^H\mathcal{S}$ above 10^{-8}, the eigenvalues below that threshold count the directions shared by two groups, and for two groups $\gamma_2 = 1 - c_F$. Because pairwise angles do not determine γ_N for $N \geq 3$ (Section 2.7), the full Gram matrix is used.

Each component u_j is then analysed again by the same procedure, up to depth two: a factor that still splits on its own component is replaced by its children. A split is accepted without further evidence if it leaves the kernel unchanged, and otherwise only if it improves the evidence of Section 4.9 by more than one nat. Complex factors of a real field are always kept together with their conjugates. The beam equation of Appendix C is an example. The core first fits it as one factor of degree two in τ, $\tau^2 + 0.0903\,\xi^4$, whose component u_1 contains, at every wavenumber, two modes with rates $\lambda = \pm 0.30\, i\kappa^2$. The recursion finds that this factor

splits along its two root branches, into the children $\tau - 0.3004\,i\xi^2$ and $\tau + 0.3004\,i\xi^2$; their kernels together are the kernel of the parent, so the split is accepted without further evidence, and the two children, which are complex conjugates, are reported together.

4.8 Identifiability and the output labels

Section 2.5 rules out the label "identified". Each reported structure carries one of four labels.

FIT-COMPATIBLE Every factor Q_j, of degree m_j in τ and d_j in ξ, has at least $m_j(d_j+1)+1$ modes (more data than coefficients), modes at d_j+1 distinct wavenumbers (Proposition 4.1), and m_j modes at some single wavenumber (one per root); and the fit passes the goodness-of-fit test below. The data are compatible with the factored equation and pin down its coefficients within the class.

ID-KER The fit is good, but some factor is determined only relative to the declared class (too few distinct wavenumbers): another equation of a larger class fits equally well.

MISFIT The goodness-of-fit test rejects the structure.

UNDECIDED The data cannot decide, typically because of confluent rates or a repeated factor.

Repeated factors ($Q = P_i^2$) are always reported as UNDECIDED: by Theorem 2.9(b) a repeated factor is visible only through polynomial prefactors $t\,e^{\lambda t}$, which finite noisy data cannot separate from two nearby rates.

4.9 Noise level and acceptance

The noise level enters in three places. The observation noise σ_n is integrated out in the interpolation (Section 4.1). Per Fourier bin it becomes σ_κ, which sets the band and the rate uncertainties. In the final decision it is used against the *raw observations*, not against the interpolated field. A candidate set of factors $Q_1, \dots, Q_N$ defines a linear model for the vector y of the n observations: let E_j be the matrix whose columns are the exponentials $e^{\lambda t + i\kappa x}$ of the kernel of Q_j (the functions spanned by B_j in Section 4.7), evaluated at the observation points, $E = [E_1 \cdots E_N]$, and let a be the vector of their amplitudes. Then

$$y = Ea + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma_n^2 I), \qquad a \sim \mathcal{N}(0, \omega^2 I),$$

where I is the identity matrix, so that $\sigma_n^2 I$ and $\omega^2 I$ are the covariance matrices of the noise and of the amplitudes, and ω^2 is the prior variance of each amplitude. Both are multivariate Gaussian distributions, and integrating the amplitudes out gives the distribution of the data under the candidate,

$$y \mid Q_1, \dots, Q_N \;\sim\; \mathcal{N}(0,\; \sigma_n^2 I + \omega^2 E E^H),$$

with E^H the conjugate transpose of E. The score of the candidate adds a complexity penalty to the logarithm of this density,

$$S(Q_1, \dots, Q_N) = \log \mathcal{N}(y;\, 0,\; \sigma_n^2 I + \omega^2 E E^H) - \tfrac{1}{2}\, n_{\mathrm{par}} \log n, \tag{23}$$

where n_{par} is the number of coefficients that define the factors. The first term is the log-probability of the observations actually made under the candidate factors, with the amplitudes integrated out; the second, as in the Bayesian information criterion, approximately accounts

for the coefficients of the factors themselves, which were fitted rather than integrated, and penalizes candidates with more of them. The score is built to *compare* candidate structures on the same data, which of two factor sets explains the observations better, not to judge one fit in isolation. It decides between candidates: dropping a factor, dropping a term whose 90% interval contains zero, merging two factors, or splitting one. A change is accepted only if it gains more than one nat. The same model gives a goodness-of-fit test: with $\hat{y}$ the projection of y on the columns of E,

$$\chi^2 = \sum_{i=1}^{n} \frac{|y_i - \hat{y}_i|^2}{\sigma_n^2},$$

which, if the candidate is right and the noise Gaussian (variance σ_n^2 per real observation), approximately follows a chi-squared distribution over noise realizations with $n - \operatorname{rank} E$ degrees of freedom; $\operatorname{rank} E$ rather than the number of columns, because a mode shared by two groups appears in both E_1 and E_2, and only approximately, because the rates in E were themselves fitted to the data. (For complex data with $\mathbb{E}|\varepsilon_i|^2 = \sigma_n^2$ the statistic is half a chi-squared variable with $2(n - \operatorname{rank} E)$ degrees of freedom.) A p-value below 10^{-3} turns the label into MISFIT. Deciding on the raw data matters: the interpolant has already smoothed the high wavenumbers, and a candidate that fits the interpolant may not fit the observations.

Finally the interaction ratio ρ of (11) is computed for every evolution factor (degree one in τ) with more than one term from the closed form (10). It is reported, not used to accept or reject: by Proposition 2.19 it is a diagnostic of the evolution inside a factor, not a test of the factor.

5 The statistical layer

The deterministic core gives one answer per field sample. The statistical layer turns the answers for all samples into one distribution. It has two parts: a Gaussian model of the components given the factors, which converts the Friedrichs angle into credible intervals, and a clustering of the factors found in all samples, which decides how many groups there are and what they are.

5.1 Gaussian prior on the components and the split layer

The core returns factors and components for one field. The first statistical question is how well the components are determined given the factors. On the observation window write

$$u = \sum_{j=1}^{N} B_j a_j + \varepsilon = \mathcal{S}a + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma_n^2 I), \qquad a \sim \mathcal{N}(0, \omega^2 I), \tag{24}$$

with B_j the orthonormal kernel bases of Section 4.7 and a_j the coefficients of component j in that basis. As in Section 4.9, ω^2 is the prior variance of each coefficient, now in the orthonormal basis: the prior covariance of a is the diagonal matrix $\omega^2 I$. This Gaussian prior on the *coefficients of the components* is the second prior of the method; the first, in Section 4.1, is on the field. Given the data, the coefficients are again Gaussian, centred at the regularized least-squares fit and with covariance Γ_a:

$$a \mid u \;\sim\; \mathcal{N}(\hat{a}, \Gamma_a), \qquad \hat{a} = \Big(\mathcal{S}^H\mathcal{S} + \frac{\sigma_n^2}{\omega^2} I\Big)^{-1} \mathcal{S}^H u, \qquad \Gamma_a = \sigma_n^2 \Big(\mathcal{S}^H\mathcal{S} + \frac{\sigma_n^2}{\omega^2} I\Big)^{-1}. \tag{25}$$

The layer therefore does not return one split of u but a Gaussian distribution of splits around the fitted one.

Proposition 5.1 (The angle sets the error bars)**.** *For two groups the eigenvalues of $\mathcal{S}^H\mathcal{S}$ are $1\pm\cos\theta_k$, with θ_k the principal angles between the sampled kernels, together with the eigenvalue 1 of multiplicity $|\dim M_1 - \dim M_2|$. In the worst direction orthogonal to $\ker\mathcal{S}$ the posterior variance of the component coefficients is*

$$\frac{\sigma_n^2}{\gamma_2 + \sigma_n^2/\omega^2}, \qquad \gamma_2 = 1 - c_F. \tag{26}$$

On the shared directions ($\theta_k = 0$, the overlap $\delta(P)$) the likelihood is flat, the posterior equals the prior, and the posterior mean splits the shared part equally.

Proof. With orthonormal B_1, B_2, $\mathcal{S}^H\mathcal{S} = \begin{pmatrix} I & B_1^H B_2 \\ B_2^H B_1 & I \end{pmatrix}$; the singular values of $B_1^H B_2$ are the $\cos\theta_k$, which gives the eigenvalues $1 \pm \cos\theta_k$ (and 1 on the directions of the larger space without a partner), and (25) is diagonal in the same basis. The smallest nonzero eigenvalue $1 - c_F$ gives (26). For $\theta_k = 0$ the likelihood depends on $a_1 + a_2$ along that direction only, so the posterior along $a_1 - a_2$ is the prior, centred at zero; its variance there is ω^2. □

Equation (26) is the Bayesian form of the stability bound (8): credible intervals grow like $\sigma_n/\sin(\theta_F/2)$ as the angle closes, and are capped by the prior when $\theta_F \to 0$. The scale ω^2 is not set by hand: it is the value that makes the observed field most probable (empirical Bayes),

$$\hat{\omega}^2 = \arg\max_{\omega^2} \ \log\mathcal{N}(u;\, 0,\ \sigma_n^2 I + \omega^2 \mathcal{S}\mathcal{S}^H),$$

maximized over a grid from 10^{-6} to 10^2. The same evidence compares the model $u \in M_1 \oplus M_2$ with the model with one factor, which gives the probability $P(\text{split})$; as $\theta_F \to 0$ the evidence ratio tends to one and $P(\text{split})$ returns to its prior. The energy shares w_j are nonlinear functions of a, so their intervals are obtained by Monte Carlo, without linearization: the layer draws 400 coefficient vectors $a^{(1)}, \ldots, a^{(400)}$ from the Gaussian (25), computes the shares of each, and reports their quantiles.

5.2 Many samples, one cloud: grouping across samples

The core runs on each of the N_s posterior field samples $u^{(q)}$ of Section 4.1. Every sample returns its own factors, with energy shares $w_j^{(q)}$. The same physical group appears once per sample, with slightly different coefficients, and sometimes split into two near-copies or merged with a neighbour. The second grouping problem is therefore to recognize the same factor across samples.

Each factor is mapped to a point of a fixed coefficient space: the monic coefficient vector $\beta \in \mathbb{C}^{m\times(d+1)}$ of its symbol, zero-padded to the largest degrees in play, so that equal operators give equal vectors whatever their position in a sample's list. The *cloud* is the weighted point measure

$$W = \sum_{q=1}^{N_\mathrm{s}} \pi_q \sum_j w_j^{(q)}\, \delta_{\beta_j^{(q)}}, \tag{27}$$

which gives each factor the mass of its sample times its energy share. Here π_q is the weight of the field sample $u^{(q)}$, inherited from the marginalization over the Gaussian-process hyperparameters (14); $w_j^{(q)}$ is the energy share (11) of factor j in that sample, computed from the components of that sample (one value per factor and sample, not an average over the Monte Carlo draws above); and δ_β is a unit point mass at the coefficient vector β, so that each factor appears in the cloud as one point. A sharp equation is a cloud with a few tight clusters; an uncertain one is a diffuse cloud.

Dirichlet-process clustering Each group of points in the cloud is modelled as a Gaussian cluster around the coefficient vector of its factor, so the cloud as a whole is described by a Gaussian mixture model (GMM): a weighted sum of Gaussian densities, one per group, with weights given by the cluster masses. A finite GMM needs the number of groups in advance. The number of groups is not known, so we use its Dirichlet-process version, which lets the data choose it: a Dirichlet-process Gaussian mixture (truncated stick-breaking with eight mixture atoms, diagonal covariances, variational inference [1]), separately for each degree m in τ. The concentration α is chosen on the grid $\{10^{-3}, \dots, 10\}$ by the evidence lower bound. Each factor enters the fit once plus a number of copies proportional to its mass; replicating every factor many times instead splits one group into several clusters. Two clusters that never occur together in the same sample are merged when a full-covariance BIC prefers one Gaussian; full covariances are needed because coefficients of one factor are strongly correlated (in the worked example of Section 6.1, the constant and the ξ^2 coefficient of the diffusion factor have correlation 0.78 across 32 samples).

The final distribution The output is a distribution, not a point estimate:

- the factors (cluster centres) with 90% intervals for every coefficient (quantiles over posterior samples, or over parametric-bootstrap refits on gridded input);
- the posterior $P(k)$ of the number k of groups, by Monte Carlo over samples and cluster responsibilities, together with the posterior over structures from (17);
- the energy shares w_j with intervals, $P(\text{split})$, γ_N, c_F and the distribution of ρ;
- the label of Section 4.8.

The effective number of groups $k_{\text{eff}} = \exp(-\sum_j \bar{w}_j \log \bar{w}_j)$, with $\bar{w}_j$ the cluster masses, is reported but never rounded to decide k: rounding is fragile (a spurious mass of 1.4% moves k_{eff} from 2.00 to 2.13), so k is read from $P(k)$.

When is the model good enough? The decision combines four quantities, in this order: the raw-data goodness of fit (otherwise MISFIT); the concentration of $P(\mathsf{M})$ on one structure (at least 0.9) and of $P(k)$ on one value; the counting condition of Proposition 4.1 (otherwise ID-KER); and the conditioning, read through γ_N and $P(\text{split})$. A small Friedrichs angle does not reject the equation, it widens the intervals of the components and lowers $P(\text{split})$. The reader therefore always sees how well each group is separated from the others, not only whether the equation fits.

6 Results

We test the method in three ways. Section 6.1 follows one data set through every stage, so that each quantity of the output can be seen and interpreted. Section 6.2 compares the method with library regression and dynamic mode decomposition on a benchmark of ten equations in three designs, and evaluates it on twenty random equations that were not used in its development.

6.1 A complete run on one data set

We follow one input through every stage. The data are the running example (Example 2.2) with $c = 1$, $\nu = 0.3$, observed on a 64×128 grid over $[0, 2] \times [0, 2\pi)$ with white noise of standard deviation $\sigma_n = 0.05\,\mathrm{std}|u|$, the hardest grid setting of the benchmark in Section 6.2. The pipeline is run exactly as in the benchmark: one noisy grid, the outer loop with default parameters, and 16 parametric-bootstrap refits for the intervals. The run takes about two minutes on one core. The figures of this subsection also use a second run on the same input with $N_\mathrm{s} = 32$ posterior samples, which shows the cloud of Section 5.2.

The band Of the 64 positive wavenumbers below $\pi/\Delta x = 64$, only $\kappa \le 4$ rise above the detection threshold (Fig. 4); the noise-free field has negligible energy beyond $\kappa \approx 5$, and a diffusive mode at $\kappa = 3$ loses a factor $e^{-0.3 \cdot 9 \cdot 2} \approx e^{-5.4}$ of its amplitude over the window. Every later decision is made with these few bins.

Modes and directions The matrix pencil returns twelve modes (Fig. 5). Eight lie on the line $\mathrm{Im}\,\lambda = -\kappa$ with $\mathrm{Re}\,\lambda \approx 0$ (transport), four on the parabola $\mathrm{Re}\,\lambda = -0.3\kappa^2$ with $\mathrm{Im}\,\lambda \approx 0$ (diffusion). Each factor is a *direction* in the frequency plane: a curve along which its modes line up. The structured fit recovers the two curves; the thin lines are the factors fitted to the 32 posterior samples and show the uncertainty of each curve, larger for diffusion at high $|\kappa|$, where there are no modes. The counting condition of Proposition 4.1 holds: the transport modes occupy eight distinct wavenumbers ($\kappa = \pm 1, \ldots, \pm 4$) and the diffusion modes four ($\pm 1, \pm 2$), against the two and three required for factors of degree one and two in ξ.

How Fig. 5 is built. Each point is one mode (κ, λ) returned by the pencil: its horizontal position is the wavenumber, its height the oscillation $\mathrm{Im}\,\lambda$ (panel a) or the damping $\mathrm{Re}\,\lambda$ (panel b), and its colour the factor to which the structured fit assigned it. Each thin line is one fitted factor, drawn as its dispersion curve $\lambda = \Lambda_j(i\kappa)$ for all κ, one line per factor and posterior sample. In the figure, the transport group is a sloped line in (a) and a flat line at zero in (b), a mode that travels without decaying; the diffusion group is flat in (a) and a downward parabola in (b), a mode that does not travel and decays faster at higher wavenumber. The thin lines bundle tightly where there are modes and fan out where there are none.

Grouping and the evidence step At the starting class the data-level evidence ranks the structure $((1,1),(1,2))$ (one factor of degree $(1,1)$, one of degree $(1,2)$) first, 0.13 nats ahead of $((1,1),(1,3))$; the second class contains the first, so both are compatible with the truth and the simpler one is kept. The core, applied to the noisy grid, initially returned *three* factors: the diffusion factor and two near-copies of the transport factor, $\tau + 1.0006\,\xi$ and $\tau + 1.0044\,\xi - 0.0028$. Grouping by evidence on the observations removed the duplicate (the factor set without it has higher evidence), and dropped the constant terms, whose intervals contained zero. An earlier

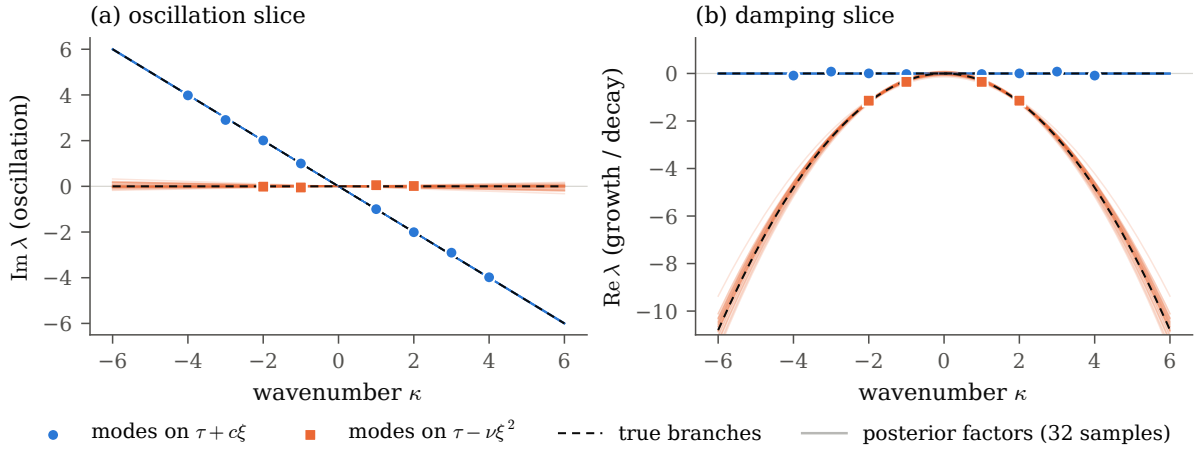


Figure 5: The two groups of the running example as directions in the frequency plane. Points: modes found by the matrix pencil in the noisy data, coloured by the factor they were assigned to. Dashed: the true dispersion branches $\lambda = -ic\kappa$ and $\lambda = -\nu\kappa^2$. Thin lines: the factors fitted to 32 posterior samples of the same input; their spread is the uncertainty of each direction. The points come from the run with the benchmark protocol, the lines from the 32-sample run. (a) Oscillation, $\mathrm{Im}\,\lambda$; (b) damping, $\mathrm{Re}\,\lambda$.

version of the pipeline (the row "previous version" of Table 5) returned a similar triple on this input, two transport copies $\tau + 0.9989\,\xi$ and $\tau + 1.0044\,\xi$ and a diffusion factor $\tau - 0.263\,\xi^2$, and labelled it FIT-COMPATIBLE: a confident wrong answer that the grouping step now prevents.

Cloud and clusters Over 32 posterior samples the core returns 64 factors. In coefficient space they form two tight clusters around the true factors (Fig. 6a), and the largest principal angle between their kernels is near zero inside a cluster and about 1.2 rad across clusters (Fig. 6b). The Dirichlet-process mixture finds two clusters with masses 0.767 and 0.233; $P(k = 2) = 1$ and $k_{\mathrm{eff}} = 1.72$ (90% interval $[1.67, 1.79]$), which is far from 2 only because the two groups carry unequal energy.

How Fig. 6 is built. In panel (a) each marker is one factor found in one posterior sample, placed at its coefficient of ξ (horizontal) and of ξ^2 (vertical), with area proportional to its mass in the cloud. Panel (b) compares every pair of these factors by the spaces of functions they allow: for each factor we take the span of its kernel exponentials on the grid and band, and compute the principal angles between the two spans from the singular values of $B_i^H B_j$ (their cosines). The colour shows the largest of these angles, light for nearly the same space and dark for very different spaces, with rows and columns ordered by cluster. The figure shows, in (a), two tight, well separated clouds and nothing in between; in (b) two light blocks on the diagonal, because all factors of one group allow the same functions, and dark blocks off the diagonal, because the two groups do not. (The largest angle is used here to tell kernels apart; the Friedrichs angle of Section 2.7 is the smallest nonzero one, which measures how well two different groups can be separated.)

Components and energy Figure 7 shows the data, the two components and the residual. The transport group carries $w_1 = 0.768$ of the energy and the diffusion group $w_2 = 0.232$, with

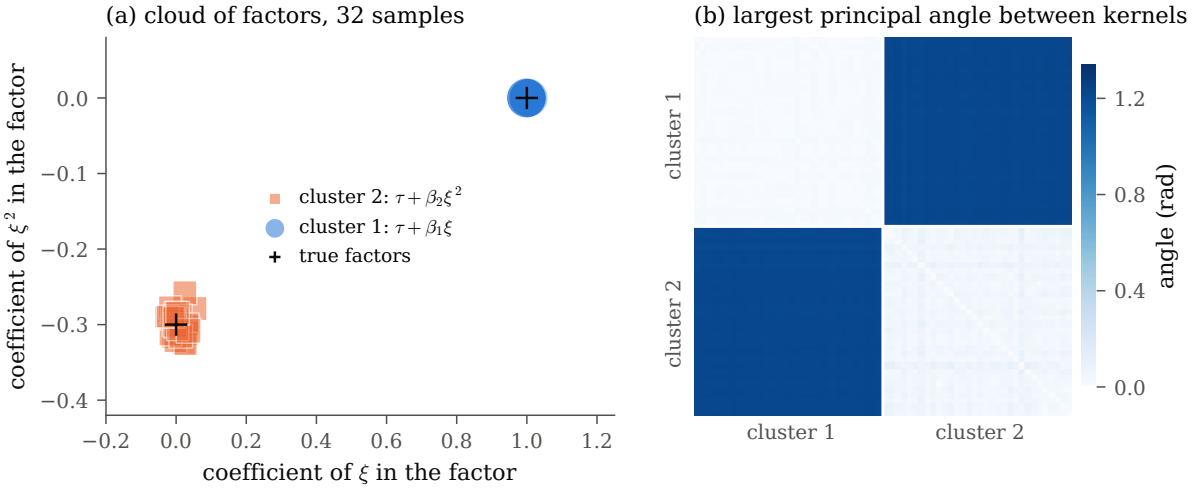


Figure 6: Grouping across posterior samples. (a) Each marker is one factor returned for one of 32 posterior samples, placed by its coefficients of ξ and ξ^2, the two that separate the groups; area is proportional to its mass $\pi_q w_j$; crosses mark the true factors. (b) Largest principal angle between the sampled kernels of every pair of factors, ordered by cluster: the block structure is the grouping.

90% intervals of width below 0.002. The joint residual of the data on the two kernels is 2.5% of the data norm, and it shows no structure (Fig. 7d). Each reported factor has a single term besides ∂_t, so its interaction ratio is $\rho = 0$ by definition. Computed on the factors of the cloud before pruning, ρ has median 0.03 (90% interval $[0.01, 0.07]$ over 32 samples), entirely due to the small constant and ξ coefficients that the evidence step removes. For comparison, on the advection–diffusion equation (one factor with the two terms $c\partial_x$ and $\nu\partial_x^2$) the same pipeline reports $\rho = 0.79$, with a 90% interval $[0.788, 0.793]$ over 16 samples: there the interaction is most of the evolution, as Theorem 2.18 predicts.

How Fig. 7 is built. Each panel is a map of a function of (t, x), with x horizontal and t vertical, red for positive and blue for negative values, on one colour scale for all panels. The components are the sums of the modes of each group, (22). The transport component consists of tilted stripes, a profile that moves at speed $c = 1$ without changing shape; the diffusion component has vertical structure that fades with time; the residual has no pattern left.

The output Table 4 lists the final output and how to read it. The separation of the two groups is moderate: $c_F = 0.827$, $\theta_F = 0.60$ rad, so errors in the components are amplified by at most $\gamma_N^{-1/2} = 2.41$. For contrast, the two conjugate factors of the beam equation $\tau \mp 0.3\, i\xi^2$ (Appendix C) are recovered with the same accuracy but are much closer to each other: $c_F = 0.939$, an amplification of 4.05.

6.2 Benchmarks

Design The benchmark has ten linear constant-coefficient equations in one space dimension: transport, heat, advection–diffusion, Airy, Schrödinger (complex field), beam, wave, Klein–Gordon, telegraph, and the running example. Each is observed in three designs: a 64×128

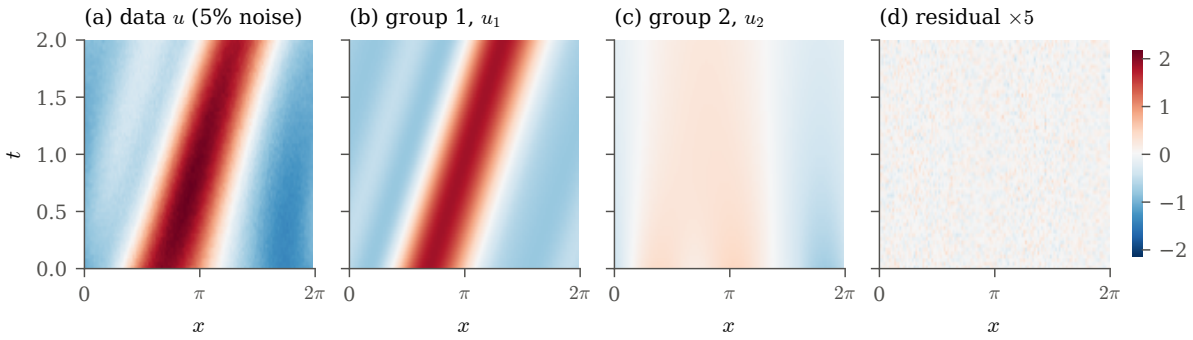


Figure 7: The data of the worked example and its two groups as functions of (t, x), on a common colour scale: (a) noisy data; (b) transport component; (c) diffusion component; (d) residual, magnified five times, which is indistinguishable from the noise.

Table 4: Output of the worked example and how to read each quantity. Intervals are 90% intervals from 16 parametric-bootstrap refits.

Quantity	Value	Meaning
equation	$(\tau + 1.0006\,\xi)(\tau - 0.2980\,\xi^2)$	recovered factors; truth $(\tau + \xi)(\tau - 0.3\,\xi^2)$, largest relative error 0.7%
coefficient of ξ	$1.0006\ [0.9998, 1.0020]$	speed c of the transport group
coefficient of ξ^2	$-0.2980\ [-0.3049, -0.2919]$	diffusivity ν of the diffusion group
label	FIT-COMPATIBLE	enough distinct wavenumbers per factor, goodness of fit passed
$P(k)$	$P(k = 2) = 1$	posterior over the number of groups
$w_1,\ w_2$	$0.768,\ 0.232$	energy shares (11) of the two groups
$P(\text{split})$	1.000	evidence that the data need both groups
$\gamma_N,\ c_F,\ \theta_F$	$0.173,\ 0.827,\ 0.60$	separation of the groups (Section 2.7)
$\gamma_N^{-1/2}$	2.41	worst-case error amplification of the components
ρ	0	no interaction inside either reported factor (single terms)

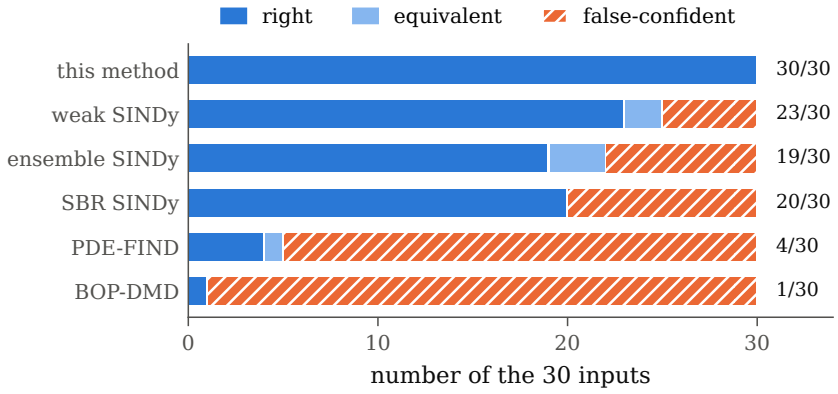


Figure 8: Outcomes on the 30 benchmark inputs (10 equations × 3 designs). Hatched: false-confident answers, i.e. a wrong equation returned without any warning.

Table 5: Benchmark outcomes per design (right or equivalent / false-confident, out of 10 each) and coefficient coverage. "Previous version" is an earlier run of the same pipeline with earlier decision rules. Runtimes are wall-clock medians per input; the pipeline ran with several workers on a shared machine.

Method	grid 1%	grid 5%	400 pts 5%	total	coef. cover$_{90}$	median time
this method	10 / 0	10 / 0	10 / 0	30 / 0	0.94	127 s
previous version	10 / 0	8 / 1	8 / 0	26 / 1	0.84	5 s
weak SINDy	10 / 0	7 / 3	8 / 2	25 / 5	–	3 s
ensemble SINDy	9 / 1	7 / 3	6 / 4	22 / 8	–	5 s
SBR SINDy	8 / 2	6 / 4	6 / 4	20 / 10	–	14 s
PDE-FIND	1 / 9	1 / 9	3 / 7	5 / 25	–	0.6 s
BOP-DMD	1 / 9	0 / 10	0 / 10	1 / 29	–	1.5 s

grid with noise 1%, the same grid with noise 5%, and 400 scattered points with noise 5% (noise standard deviation relative to std$|u|$). This gives 30 inputs. We compare with weak SINDy, ensemble SINDy, SBR SINDy [13, 6, 8], PDE-FIND [17] and BOP-DMD [18], each run on the same input (on the scattered designs, on the same interpolated field) through the reference implementations PySINDy and PyDMD. The SINDy variants use a linear library of mixed derivatives, a target $\partial_t^m u$ of every order the benchmark needs, and a sparsity threshold selected by BIC over a grid of fourteen values (zero and $10^{-4}, \ldots, 1$); their answers are therefore tuned to the same linear class as ours. An answer is *right* if the recovered set of factors matches the true one within relative coefficient error 0.05; *equivalent* if it is the same product polynomial written differently; an *abstention* if the method declines (labels ID-KER or UNDECIDED); and *false-confident* if it returns a wrong equation as if it were right. The baselines have no way to abstain, so every wrong baseline answer is false-confident.

Results The method returns the right factor set on all 30 inputs and is never false-confident (Fig. 8 and Table 5). The best baseline, weak SINDy, is right or equivalent on 25 and false-confident on 5: the beam equation and the running example in two designs each, and the Schrödinger equation on the grid at 5% noise. Four of the five are products of factors (the

Table 6: Grid designs repeated with independent fields and noise (seeds 1–3): right / false-confident out of 30 per noise level. The weak-SINDy rows were rerun here with PySINDy 2.1.0 and the settings of Table 5.

Method	grid 1%	grid 5%	total
this method	30 / 0	30 / 0	60 / 0
weak SINDy	26 / 4	21 / 9	47 / 13

wave equation, also a product, is handled correctly by weak SINDy); the other SINDy variants fail on the same products and on further inputs. Where the method returns intervals, the true coefficients fall inside the 90% intervals in 94% of cases, and the true energy shares fall inside their intervals in all six inputs where a split was reported. The comparison with the previous version isolates the effect of the grouping decisions of Section 4.9: its one false-confident answer is the duplicated transport factor of Section 6.1.

Independent repetitions Because the 30 inputs above share one field and one noise draw per equation, we repeated the two grid designs with three further seeds, each drawing a new field *and* a new noise realization: 10 equations × 2 noise levels × 3 seeds, 60 independent inputs. Table 6 compares the method with weak SINDy, the strongest baseline of Table 5. The method returns the right factor set with the label FIT-COMPATIBLE on all 60 inputs, with coefficient coverage 0.94 at nominal 0.90; weak SINDy is false-confident on 13, again mostly on products of factors (the running example 5 times, the beam twice) and on the Schrödinger equation. With 60 independent trials and no failure, a one-sided 95% upper bound on the method's false-confident rate on these designs is 5%.

Equations outside the benchmark list To test generalization we drew 20 random equations from five families: first-order factors (F1), first-order dispersive factors (F2), irreducible second-order factors (F3), products of two or three factors (F4), and third-order factors or a reaction factor times a transport factor, often with few excited wavenumbers (F5). None of these draws was used to set any parameter. Each is observed on the grid with 1% noise. Table 7 reports the outcomes. "Equivalent" here means that the returned equation explains the data as well as the true one by a χ^2 test adjusted for degrees of freedom, and "abstain-ok" that the method abstained on an input that is genuinely not resolvable at this noise level: a factor is resolvable when, computed from the truth alone, it has modes above the detection threshold at $d_j + 1$ distinct $|\kappa|$ (a stricter count than the label rule of Section 4.8) and each of its terms moves some visible rate by more than three standard deviations.

The core alone is false-confident on six of the twenty draws, the full method on none: the outer loop and the statistical layer, not the spectral fit, are what turn a fit into a trustworthy answer. Five answers carry an abstaining label although their factors are right (two) or equivalent (three); they are counted by their factors, and no law is claimed for them.

What these numbers do and do not show All inputs are synthetic and noise is white. The 30 benchmark inputs share one noise draw per equation, so they are 10 equations in three correlated designs, not 30 independent trials; Table 6 supplies independent repetitions for the grid designs, but not for the scattered design, whose Gaussian-process interpolation we did

Table 7: Twenty random equations from five families, grid, 1% noise. "Core" is the deterministic core of Sections 4.4 and 4.5 run once, without the outer loop and the statistical layer.

Method	right	equivalent	abstain-ok	abstain	false-confident
this method	12	7	1	0	0
core only	5	3	3	3	6
ensemble SINDy	6	1	0	0	13
weak SINDy	5	1	0	0	14
PDE-FIND	2	0	0	0	18

not rerun. The default parameters were tuned during development on test inputs built from the same ten equations, so Table 5 is not a held-out evaluation; Table 7 is. Runtimes are not comparable across methods: the method fits 90 structures at the starting class (1353 if the class grows), each on every sample, and runs a bootstrap. The claims we draw are therefore qualitative: on equations whose structure is a product of factors, grouping by factors recovers the structure where regression on a term library does not, and the labels and intervals warn instead of returning a wrong equation.

7 Inference problems: solvable and unsolvable

The statistical layer inherits limits from the analysis. Some are engineering problems with a known remedy; others are theorems and no amount of data or modelling removes them.

Problems with a remedy

- *Choosing k.* Rounding k_{eff} is unstable; the posterior $P(k)$ replaces it.
- *Correlated noise after interpolation.* Residuals of a Gaussian-process interpolant are correlated, so white-noise likelihoods miscalibrate the intervals. The remedy is generalized least squares per Fourier bin, with the bin covariance estimated from the posterior draws; the angle should then be measured in the inner product of the inverse posterior covariance, which the method approximates bin by bin.
- *Matching factors across samples.* A nearest-vector match without a gate merged transport and heat; Dirichlet-process clusters with a co-occurrence merge rule fixed it.
- *Real fields.* A complex-valued component model fitted to a real field undercovered; a real parametrization by conjugate pairs restored nominal coverage.

Problems without a remedy

- *Error floor.* In the Gaussian model (24) with a flat prior, every unbiased estimator of the component coefficients has variance at least σ_n^2/γ_N along the worst direction (the Cramér–Rao bound: σ_n^2 times the inverse of the smallest nonzero eigenvalue of $\mathcal{S}^H\mathcal{S}$, Proposition 5.1), and the minimum-norm split attains the corresponding amplification $\gamma_N^{-1/2}$ (Theorem 2.17): closely aligned groups are intrinsically hard to separate.
- *No data-free threshold.* Whether a numerical symbol "factors" depends on the noise level; there is no threshold independent of the data, which is why the decision is by evidence and not by a tolerance on coefficients.

Table 8: The kinds of claim made in the paper.

Kind	Statements
proved	groups are factors with overlap $\delta(P)$ (Theorem 2.14); terms are not groups (Propositions 2.16 and 2.19); identification criterion and its limits on finite data (Theorem 2.9 and Propositions 2.10 and 2.13); stability via γ_N (Theorem 2.17); Bayesian error bars via the angle (Proposition 5.1); counting condition (Proposition 4.1)
design rule	labels instead of "identified"; distributions of ρ instead of values; decisions on the raw observations; growth of the class instead of coefficient penalties; minimum-norm split of the overlap
measured	0/20 confident wrong answers on random equations not used in development; 60/60 factor sets on grid designs with independent fields and noise; 30/30 on the development benchmark; coefficient coverage 0.94 at nominal 0.90

- *Repeated factors.* Finite noisy data cannot distinguish P_i^2 from two nearby factors; the label is UNDECIDED.
- *Loss of a derivative.* Components may be one derivative rougher than u; the prior on the components must be one order smoother than the prior on the field.
- *No certificate.* By Section 2.5 no finite data certify an equation; outputs are compatibility statements with uncertainty.

8 Conclusions

The factorization of the symbol of a linear constant-coefficient PDE is a grouping structure on its solutions: the modes of one factor lie on one curve of the dispersion relation, the solution splits into one component per factor up to an overlap of dimension $\delta(P)$, and the components are as well separated as the Friedrichs angle between their solution spaces allows. Additive terms do not form groups, and their interaction cannot be regularized away. These facts determine what a learning algorithm should look for and what it can promise; Table 8 separates what is proved from the design rules derived from it and from what is measured.

The algorithm built on them groups twice, modes into factors within a sample and factors across samples, and reports a distribution over equations whose labels and intervals reflect the limits proved in Section 2: an equation is never certified, closely aligned groups come with wide intervals, and repeated factors are reported as undecided. On twenty random equations not used in development it is never wrong without warning; on the thirty-input development benchmark it recovers every factor set, where the best regression on a library of terms fails without warning on five inputs, four of them products of factors.

Three directions remain open. For variable coefficients the eigenvalues of the pencil become functions of position (Remark 4.2), and the factors become families of curves; the algebra is then non-commutative, and a splitting theorem is not available. For nonlinear equations the ring is replaced by a dictionary and grouping must be relative to it (Section 3). For systems such as the Dirac equation the ideal becomes a module and the conditions for splitting become topological. In each case the question asked in this paper, which pieces of the equation a solution carries and how stably, is the right starting point.

A Proofs

Throughout, $\Omega \subset \mathbb{R}^2$ is open and connected, $\mathcal{A}$ is $\mathcal{D}'(\Omega)$ or $C^\infty(\Omega)$, and $\mathcal{R} = \mathbb{C}[\tau, \xi]$ acts on $\mathcal{A}$ by $\tau \mapsto \partial_t$, $\xi \mapsto \partial_x$. For $z = (\tau_z, \xi_z) \in \mathbb{C}^2$, $\mathfrak{m}_z = (\tau - \tau_z, \xi - \xi_z)$ is its maximal ideal and $\psi_z(t, x) = e^{\tau_z t + \xi_z x}$.

A.1 Algebraic toolkit

Proof of Lemma 2.4. Coprime polynomials meet in finitely many points. Let $f, g \neq 0$ be coprime. By Gauss's lemma they are coprime in the principal ideal domain $\mathbb{C}(\xi)[\tau]$, so $af + bg = 1$ there; clearing denominators, $a'f + b'g = r(\xi) \neq 0$ with $a', b' \in \mathcal{R}$. At a common zero $r(\xi_0) = 0$, which leaves finitely many ξ_0; exchanging the variables, finitely many τ_0. Hence $V(f, g)$ is finite.

Splitting. Put $J = \{f/g : \ f \in I\}$ with $g = \gcd(I)$; it is an ideal and $I = gJ$, and no prime divides all of J. If $J \neq \mathcal{R}$, pick $0 \neq f \in J$ with distinct prime factors $\pi_1, \dots, \pi_r$ and, for each a, $g_a \in J$ with $\pi_a \nmid g_a$. The sets $\{c \in \mathbb{C}^r : \pi_a \mid \sum_b c_b g_b\}$ are proper linear subspaces, so some c avoids all of them; then f and $\sum_b c_b g_b \in J$ are coprime and $V(J) \subseteq V(f, \sum_b c_b g_b)$ is finite. By the Nullstellensatz a power of $\sqrt{J}$ lies in J, and $J = \mathcal{R}$ when $V(J) = \emptyset$. □

Proof of Lemma 2.7. ($\Rightarrow$) $Q(\partial)(q\,\psi_z) = (Q(z + \partial)q)\psi_z$ for every polynomial q, so the derivatives of $q\,\psi_z$ stay in the finite-dimensional space of polynomials of degree $\leq \deg q$ times ψ_z. Moreover $\prod_k \mathfrak{m}_{z_k}^{\deg q_k + 1} \subseteq \operatorname{Ann}(w)$, so $V(\operatorname{Ann} w) \subseteq \{z_k\}$; conversely $Q \in \operatorname{Ann}(w)$ forces $Q(z_k + \partial)q_k = 0$ by the linear independence of distinct exponentials, whose top-degree part is $Q(z_k)$ times that of q_k, so $Q(z_k) = 0$.

($\Leftarrow$) Let $\mathcal{R}w$ be finite-dimensional and p, q the minimal polynomials of ∂_t and ∂_x restricted to $\mathcal{R}w$, so that $p(\partial_t)w = 0$ and $q(\partial_x)w = 0$. On an open rectangle $I' \times J' \subset \Omega$, induction on $\deg p$ shows that every distribution with $p(\partial_t)v = 0$ is $\sum_{j,l} c_{jl}(x)\, t^l e^{\lambda_j t}$ with $c_{jl} \in \mathcal{D}'(J')$ and λ_j the roots of p (the base case uses that a distribution with $\partial_t V = 0$ on a rectangle is $c(x) \otimes 1$). Applying $q(\partial_x)$ and testing against functions dual to $t^l e^{\lambda_j t}$ gives $q(\partial_x)c_{jl} = 0$, so each c_{jl} is an exponential-polynomial of x. Thus w is an exponential-polynomial E near every point. The set where w coincides near y with a fixed E is open, and it is closed because two exponential-polynomials that agree on an open set agree everywhere; by connectedness it is Ω. □

A.2 Identification

Proof of Theorem 2.9. Let $I = \operatorname{Ann}(u) \ni P$ and write $I = G\,J$ as in Lemma 2.4. Since $P \in I$, $G \mid P$, so $G = c' \prod_i P_i^{f_i}$ with $0 \leq f_i \leq e_i$. First, $G \sim P$ iff $I = (P)$: if $G \sim P$ then $I = PJ \subseteq (P) \subseteq I$; if $I = (P)$, P divides all of I and lies in I.

(a), $\Leftarrow$ by contraposition. If $I \neq (P)$, some $f_i < e_i$, so $P/P_i = G\,H$ with $H \in \mathcal{R}$. A power $\mathfrak{n}^\ell$ of the ideal $\mathfrak{n}$ of the finite set $V(J)$ lies in J, and for $h \in \mathfrak{n}^\ell$, $h \cdot P/P_i = G(Hh) \in GJ = I$, i.e. $h(\partial)v_i = 0$. So $V(\operatorname{Ann} v_i) \subseteq V(J)$ is finite and $v_i \in \mathrm{EP}$ by Lemma 2.7.

(a), $\Rightarrow$ by contraposition. If $v_i \in \mathrm{EP}$, $V(\operatorname{Ann} v_i)$ is finite while $V(P_i)$ is infinite, so some $h \in \operatorname{Ann}(v_i)$ is not divisible by P_i. Then $Q = h \cdot P/P_i$ satisfies $Q(\partial)u = h(\partial)v_i = 0$; if $Q = PA$, cancelling P/P_i gives $h = P_i A$, a contradiction. So $Q \in I \setminus (P)$.

(b) $V(I) = V(G) \cup V(J)$ with $V(J)$ finite. If $f_i \geq 1$, $V(P_i) \subseteq V(I)$; if $f_i = 0$, $V(P_i) \cap V(I)$ lies in the finite set $\bigcup_{f_j \geq 1} V(P_i, P_j) \cup V(J)$, so $V(P_i) \not\subseteq V(I)$. Hence $V(I) = V(P)$ iff all $f_i \geq 1$, and $f_i = 0$ iff $v_i' \in \mathrm{EP}$ by the argument of (a) with $P/P_i^{e_i}$ in place of P/P_i.

(c) If $e_i = 1$ then $v_i = v_i'$. If P is monic of degree m in τ, divide any $Q \in \mathcal{R}$ by P in $\mathbb{C}[\xi][\tau]$: $Q = AP + \sum_{j<m} \varphi_j(\xi)\tau^j$ uniquely. Then $Q \in I$ iff $\sum_j \varphi_j(\partial_x)\partial_t^j u = 0$, and $Q \in (P)$ iff all $\varphi_j = 0$. □

Proof of Proposition 2.10. (a) If $I = (P)$, every annihilator is SP and $\deg SP \geq \deg P$ with equality iff S is a nonzero constant. (b) If $I = (P)$ then $G \sim P$ and $J = \mathcal{R}$, so $\Sigma_{\mathrm{p}}(u) = \emptyset$. Conversely, $V(J) = \emptyset$ gives $J = \mathcal{R}$ by the Nullstellensatz, hence $I = (G) = (P)$. (c) Let f_i be the multiplicity of P_i in G and $w = (P/P_i^k)(\partial)u$. If $k \leq e_i - f_i$, then $G \mid P/P_i^k$, say $P/P_i^k = GH$; for h in a power $\mathfrak{n}^\ell \subseteq J$ of the ideal of $V(J)$, $h \cdot P/P_i^k = G(Hh) \in I$, so $\mathrm{Ann}(w) \supseteq \mathfrak{n}^\ell$ and $w \in \mathrm{EP}$. If $k > e_i - f_i$ and $w \in \mathrm{EP}$, pick $h \in \mathrm{Ann}(w)$ with $P_i \nmid h$ (as in the proof of Theorem 2.9); then $hP/P_i^k \in I$, so $G \mid hP/P_i^k$, but P_i occurs in hP/P_i^k with multiplicity $e_i - k < f_i$, a contradiction. Hence $f_i = e_i - \max\{k : \ w \in \mathrm{EP}\}$. □

A.3 Decomposition and stability

Proof of Theorem 2.14. We prove the statement for open connected Ω with $H^1_{dR}(\Omega) = 0$, which holds for simply connected domains.

Step 1: finite modules have no obstruction. Let M be an $\mathcal{R}$-module of finite dimension over $\mathbb{C}$. Then $\mathrm{Ext}^1_{\mathcal{R}}(M, \mathcal{A}) = 0$ and $\dim \mathrm{Hom}_{\mathcal{R}}(M, \mathcal{A}) = \dim M$. By the long exact sequence it is enough to check one-dimensional modules $\mathcal{R}/\mathfrak{m}_z$, since M has a composition series with such quotients. A free resolution of $\mathcal{R}/\mathfrak{m}_z$ is the Koszul complex of $(\tau - \tau_z, \xi - \xi_z)$, so $\mathrm{Ext}^\bullet(\mathcal{R}/\mathfrak{m}_z, \mathcal{A})$ is the cohomology of $\mathcal{A} \to \mathcal{A}^2 \to \mathcal{A}$, $v \mapsto \big((\partial_t - \tau_z)v, (\partial_x - \xi_z)v\big)$, $(a, b) \mapsto (\partial_t - \tau_z)b - (\partial_x - \xi_z)a$. Multiplication by ψ_{-z} turns it into the de Rham complex of Ω with coefficients in $\mathcal{A}$, whose first cohomology is $H^1_{dR}(\Omega) = 0$ (for distributions by de Rham's theorem), and whose zeroth cohomology is the constants, of dimension one.

Step 2: two factors. For coprime Q_1, Q_2, $(Q_1) \cap (Q_2) = (Q_1 Q_2)$ by unique factorization, and $0 \to \mathcal{R}/(Q_1Q_2) \to \mathcal{R}/(Q_1) \oplus \mathcal{R}/(Q_2) \to \mathcal{R}/(Q_1, Q_2) \to 0$ is exact ($1 \mapsto (1, 1)$, $(a, b) \mapsto a - b$). Applying $\mathrm{Hom}_{\mathcal{R}}(\cdot, \mathcal{A})$, and using $\mathrm{Hom}(\mathcal{R}/(f), \mathcal{A}) = \ker_{\mathcal{A}} f$, gives the exact sequence $0 \to \mathrm{Hom}(\mathcal{R}/(Q_1, Q_2), \mathcal{A}) \to \ker Q_1 \oplus \ker Q_2 \xrightarrow{(u_1,u_2)\mapsto u_1+u_2} \ker Q_1 Q_2 \to \mathrm{Ext}^1(\mathcal{R}/(Q_1, Q_2), \mathcal{A})$. The quotient $\mathcal{R}/(Q_1, Q_2)$ is finite-dimensional by Lemma 2.4, so by Step 1 the last group vanishes: the sum map is onto, and its kernel has dimension $\dim \mathcal{R}/(Q_1, Q_2)$ and consists of solutions of a zero-dimensional ideal, i.e. of exponential-polynomials.

Step 3: induction. $P' = Q_2 \cdots Q_N$ is coprime to Q_1, so $\ker P = \ker Q_1 + \ker P'$, and by induction $\ker P' = \sum_{j \geq 2} \ker Q_j$. Dimensions of kernels add along the composition of the two sum maps, and $\dim \mathcal{R}/(Q_1, P') = \sum_{j \geq 2} \dim \mathcal{R}/(Q_1, Q_j)$ because $\mathcal{R}/(Q_1, Q_2Q_3 \cdots)$ is the direct sum of its localizations at the finitely many common zeros, where the intersection multiplicities add. This gives (5). □

Proof of Theorem 2.17. The sum operator $\mathcal{S}$ is bounded from $\bigoplus_j M_j$ into $\ker_{H^s} P$, and every u in its range has components in H^s. Its range is all of $\ker_{H^s} P$ iff the range is closed and dense. By the closed range theorem the range is closed iff $\mathcal{S}$ is bounded below on $(\ker \mathcal{S})^\perp$, i.e. iff $\gamma_N > 0$, and then the minimum-norm preimage satisfies (8). For two closed subspaces, $M_1 + M_2$ is closed iff $c_F < 1$, and $\gamma_2 = 1 - c_F$ [4]. In finite dimension every subspace sum is closed, and the sum equals the span of the sampled exponentials of P, each of which lies in the kernel of one factor. □

Proof of Proposition 2.19. For real $\zeta > c/\nu$ let $u_\zeta = e^{\lambda_\zeta t + \zeta x}$, $\lambda_\zeta = \nu\zeta^2 - c\zeta > 0$; then $Q(\partial)u_\zeta = 0$ and $\|u_\zeta\|_{H^s(\Omega)} \le C_s(1+\zeta)^{2s} e^{\lambda_\zeta T}\|e^{\zeta x}\|_{L^2(a,b)}$. With base time t_0 and horizon t, Theorem 2.18 gives

$$E_{\text{int}} = e^{\lambda_\zeta t_0}\big(e^{-c\zeta t} - 1\big)\big(e^{\nu\zeta^2 t} - 1\big)\, e^{\zeta x},$$

whose $L^2(\Omega)$ norm is at least a constant times $e^{\nu\zeta^2 t}\|e^{\zeta x}\|_{L^2(a,b)}$ for large ζ. With $t_0 = 0$ and $t = T$ the ratio therefore grows at least like

$$e^{\nu\zeta^2 T - \lambda_\zeta T}(1+\zeta)^{-2s} = e^{c\zeta T}(1+\zeta)^{-2s},$$

which is unbounded for every s. For the second statement, $v + w$ is killed by Q and by $R = (\tau + c\xi)(\tau - \nu\xi^2)$. At a common zero of Q and R, either $\tau = -c\xi$, and then $Q = 0$ gives $\nu\xi^2 = 0$, or $\tau = \nu\xi^2$, and then $c\xi = 0$; in both cases $(\tau, \xi) = (0, 0)$. So (Q, R) has a single zero, and its common kernel is a finite-dimensional space of exponential-polynomials (Lemma 2.7). □

B Algorithm and parameters

Algorithm 1 Grouping-based identification of a constant-coefficient PDE

Require: observations $\{(t_i, x_i, y_i)\}$ or a noisy grid; noise model; starting class $(m, d) = (3, 4)$
draw N_{s} posterior field samples $u^{(q)}$ with weights π_q ▷ Section 4.1
repeat ▷ outer loop, Section 4.3
  for each sample $u^{(q)}$ **do**
    FFT in x; keep the band $\mathcal{K}$ above $z\,\sigma_\kappa$ and below both Nyquist limits ▷ Section 4.2
    matrix pencil per $\kappa \in \mathcal{K}$: modes $(\kappa, \lambda_p, A_p, \sigma_{\lambda,p})$ ▷ Section 4.4
    for each structure $\mathsf{M} \in \mathcal{C}(m, d)$ **do**
      k-varieties fit: assign modes to factors and fit the factors ▷ Section 4.5
    end for
    $P(\mathsf{M}) \propto e^{-\text{BIC}/2}$; components u_j, shares w_j, γ_N, labels ▷ Sections 4.6 and 4.8
    recurse on each component (depth ≤ 2) ▷ Section 4.7
  end for
  pool over samples; if most mass is MISFIT or ID-KER, propose a larger class
until the larger class fails the held-out check, or $(m, d) = (4, 8)$
regroup: refit every sample at the consensus structure
cloud W of factor vectors; Dirichlet-process clustering; merge non-co-occurring clusters ▷ Section 5.2
split layer: posterior of the components, $P(\text{split})$, intervals of w_j ▷ Section 5.1
evidence on the raw observations: drop, prune, merge or split factors; goodness of fit ▷ Section 4.9
return factors with intervals, $P(k)$, $P(\mathsf{M})$, w_j, γ_N, c_F, distribution of ρ, label

C Further cases

The same protocol as in Section 6.1 (grid, 5% noise, bootstrap intervals) on four more equations of the benchmark. All are recovered with the label FIT-COMPATIBLE.

Table 9: Parameters a user may need, with their defaults. All other settings are internal and fixed.

Parameter	Default	Meaning
starting class (m, d)	$(3, 4)$	total degree in τ and degree in ξ of the first class of structures
class limits	$(4, 8)$	largest class the outer loop may grow to
concentration of $P(\mathsf{M})$	0.9	posterior mass on one structure required to stop growing
growth test level	$\log_{10} p = -3$	level of the goodness-of-fit test that triggers growth of the class
detection threshold z	3	a Fourier bin is used only above $z\,\sigma_\kappa$
misfit cap	3	a mode farther than 3 standard deviations from all varieties costs a fixed 9
evidence margins	1; $\log 150$	log Bayes factor needed to change the factor set; "very strong" level [11]
posterior samples N_{s}	8	field samples passed to the core
bootstrap refits	16	refits for coefficient intervals on gridded input
coefficient draws	16	Monte Carlo draws for the intervals of the energy shares
DP concentration grid	$10^{-3}, \dots, 10$	candidates for α, chosen by the evidence lower bound
prior scale grid	$10^{-6}, \dots, 10^{2}$	candidates for ω^2, chosen by empirical Bayes

Table 10: Further single runs. "Error" is the largest relative coefficient error.

Equation	Recovered	Groups	Separation	Error
beam	$(\tau - 0.3004\, i\xi^2)(\tau + 0.3004\, i\xi^2)$	2 (conjugate)	$c_F = 0.939$, $\gamma_N^{-1/2} = 4.05$	0.14%
Klein–Gordon	$\tau^2 - 0.9997\,\xi^2 + 0.9996$	1 (irreducible)	–	0.04%
telegraph	$\tau^2 + 0.8008\,\tau - 0.9990\,\xi^2$	1 (irreducible)	–	0.10%
advection–diffusion	$\tau + 1.0002\,\xi - 0.3004\,\xi^2$	1 (two terms)	$\rho = 0.79$	0.14%

The beam and the running example are the two inputs of the benchmark on which every SINDy variant fails at 5% noise. Both are products of factors; the beam's factors are complex conjugates and close to each other, so its components carry the largest error amplification of all cases. Klein–Gordon and telegraph are irreducible: the method returns one group and does not split them, avoiding the spurious splits produced by fitting a polynomial and factoring it afterwards (Section 4.5).

D Reproducibility

The code (package `pdefactor`), the benchmark scripts and the scripts that produce every number and figure of this paper are available in the accompanying repository. The figures are produced from stored run outputs by `figures/make_figs.py`; the runs by `experiments/worked_example.py` (benchmark protocol) and the report script of the package, and the data behind the figures by `experiments/extract_data.py`. The independent repetitions of Table 6 are produced by `experiments/multiseed.py`; their per-input results are stored with the paper. Each run takes between ten seconds and five minutes on one core. The benchmark tables are generated by the benchmark's report script from the stored per-input results.

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